

# NYC Mobility Pipeline and Insights

## Taxi Zone Prioritisation for Mobility Interventions

Thuy Linh Dang | 21878037

BUS2003 Data Engineering and Python

Assessment 3: Data Engineering Project



## Executive Summary

This report builds a reproducible data pipeline to analyse NYC mobility data and identify which taxi zones should be prioritised for targeted mobility interventions. Yellow and green taxi trip-records were used as the anchor datasets, supported by taxi zone lookup data, subway hourly ridership, bus hourly ridership and Citi Bike trip data.

The pipeline sourced, stored, audited, cleaned and transformed raw public datasets into engineered summary tables for analysis. The data story shows that taxi demand is unevenly distributed across geography and time, with different areas showing different mobility signals. Some zones show high taxi volume, some show stronger late-night reliance, some show short-trip concentration, and others show longer distance travel patterns.

The report recommends that NYC use targeted interventions rather than one city wide response. High-volume zones should be reviewed for taxi pickup/drop-off management, late-night taxi reliant zones for public transport support, short-trip zones for bike or bus alternatives, and long distance zones for connectivity review. These recommendations are presented as candidate areas for further planning investigation, not final infrastructure decisions.

# Table of Contents

|                                                                                  |                  |
|----------------------------------------------------------------------------------|------------------|
| <b><i>I. Introduction and Business Question.....</i></b>                         | <b><i>5</i></b>  |
| <b><i>II. Data Sources and Scope .....</i></b>                                   | <b><i>6</i></b>  |
| <b><i>III. Data Engineering Pipeline .....</i></b>                               | <b><i>7</i></b>  |
| a) Data sourcing and Loading.....                                                | 7                |
| b) Raw data audit .....                                                          | 10               |
| c) Cleaning and transformation.....                                              | 13               |
| d) SQL analysis and engineered summary tables .....                              | 16               |
| <b><i>IV. Data Governance, Privacy and Ethics.....</i></b>                       | <b><i>19</i></b> |
| <b><i>V. Data Story: Analysis and Findings .....</i></b>                         | <b><i>21</i></b> |
| a) Overview: taxi demand is concentrated by place and time .....                 | 21               |
| b) Issue identification: taxi patterns reveal different mobility needs .....     | 25               |
| c) Recommendations: intervention priorities should match the demand signal ..... | 29               |
| <b><i>VII. Conclusion .....</i></b>                                              | <b><i>32</i></b> |
| <b><i>References .....</i></b>                                                   | <b><i>33</i></b> |
| <b><i>Appendices .....</i></b>                                                   | <b><i>34</i></b> |
| <b><i>Appendix A: Raw Data Schema and Sample Records .....</i></b>               | <b><i>34</i></b> |
| Appendix A.1 Taxi Zone Lookup Schema and Sample Records.....                     | 34               |
| Appendix A.2 Subway Schema and Sample Records.....                               | 35               |
| Appendix A.3 Bus Schema and Sample Records.....                                  | 36               |
| Appendix A.4 Citi Bike Schema and Sample Records .....                           | 36               |
| <b><i>Appendix B: Audit Severity Classification Rules.....</i></b>               | <b><i>37</i></b> |
| <b><i>Appendix C: Supporting Dataset Audit Results.....</i></b>                  | <b><i>37</i></b> |
| Appendix C.1 Taxi Zone Lookup Audit Summary .....                                | 37               |
| Appendix C.2 Subway Audit Summary.....                                           | 38               |
| Appendix C.3 Bus Audit Summary.....                                              | 38               |
| Appendix C.4 Citi Bike Audit Summary .....                                       | 39               |
| <b><i>Appendix D: Cleaning and Transformation Results.....</i></b>               | <b><i>39</i></b> |

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| Appendix D.1 Taxi Feature Engineering Results .....             | 39        |
| Appendix D.2 Subway Cleaning Results.....                       | 40        |
| Appendix D.3 Bus Cleaning and Route-Proxy Results .....         | 40        |
| Appendix D.5 Citi Bike Cleaning Results .....                   | 41        |
| <b>Appendix E: SQL Summary Tables .....</b>                     | <b>41</b> |
| Appendix E.1 taxi_borough_summary.....                          | 41        |
| Appendix E.2 taxi_hourly_summary.....                           | 42        |
| Appendix E.3 taxi_zone_summary.....                             | 42        |
| Appendix E.4 Short-Trip Zone Summary.....                       | 43        |
| Appendix E.5 Long-Trip Zone Summary.....                        | 43        |
| Appendix E.6 taxi_subway_comparison.....                        | 44        |
| Appendix E.7 taxi_bus_comparison.....                           | 45        |
| Appendix E.8 taxi_bike_hourly_comparison .....                  | 45        |
| <b>Appendix F: Data Ethics and Governance Checkpoints .....</b> | <b>47</b> |
| Appendix F.1 Data Minimisation Log.....                         | 47        |
| Appendix F.2 Aggregation and Privacy Control Log .....          | 47        |
| Appendix F.3 Responsible Wording Log .....                      | 47        |
| <b>Appendix G: Taxi Demand Overview Outputs .....</b>           | <b>47</b> |
| Appendix G.1 Borough-Level Taxi Summary .....                   | 47        |
| Appendix G.2 Pickup Zone Summary and Ranking .....              | 48        |
| Appendix G.3 Taxi Hourly Demand Summary .....                   | 48        |
| <b>Appendix H: Detailed Taxi Pattern Outputs .....</b>          | <b>49</b> |
| Appendix H.1 Detailed Distance-Pattern Outputs .....            | 49        |
| Appendix H.2 Detailed Late-Night Zone Outputs.....              | 49        |



# I. Introduction and Business Question

New York City has a complex multi-modal transport system that includes subway, bus, taxis, for hire vehicles and bike-share services (New York City Department of Transportation, 2019). While subway and bus services carry the largest passenger volumes, taxis remain important because they provide flexible point to point transport for trips that may be less convenient, less accessible or less direct by fixed-route public transport (Harvard Kennedy School, 2014).

The main decision question for this report is:

*Which NYC taxi zones should be prioritised for targeted mobility interventions, based on taxi demand volume, time-of-day reliance, trip distance patterns, and comparison with subway, bus and Citi Bike activity in April 2026?*

The intended audience is a **New York City mobility or transport planning Department**.

The analysis supports an early-stage planning decision: identifying taxi zones that may require further service planning review. It does not make final infrastructure decisions.

## II. Data Sources and Scope

This project uses publicly available New York City transport datasets for April 2026, with a geographic scope covering NYC taxi zones and boroughs. NYC TLC yellow and green taxi trip records are used as the anchor datasets, supported by taxi zone lookup data, MTA subway hourly ridership, MTA bus hourly ridership and Citi Bike trip data.

| Dataset                                              | Role in analysis                                                                                  | Key fields used                                                                                                                  |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Yellow taxi trip records, April 2026</a> | Measures core taxi demand, timing, distance, cost, and locations.                                 | Pickup/drop-off datetime, trip distance, <a href="#">fare</a> amount, total amount, passenger count, pickup/drop-off location ID |
| <a href="#">Green taxi trip records, April 2026</a>  | Adds outer-borough street-hail taxi activity.                                                     | Pickup/drop-off datetime, trip distance, <a href="#">fare</a> amount, total amount, passenger count, pickup/drop-off location ID |
| <a href="#">Taxi zone lookup</a>                     | Links taxi location IDs to boroughs and zones for geographic analysis.                            | <del>LocationID</del> , Borough, Zone, Service Zone                                                                              |
| <a href="#">Subway hourly ridership, April 2026</a>  | Compares taxi activity with subway ridership patterns.                                            | Transit timestamp, station complex, borough, ridership, payment/fare category                                                    |
| <a href="#">Bus hourly ridership, April 2026</a>     | Compares taxi activity with bus ridership, especially for short-trip alternatives.                | Transit timestamp/hour, bus route, borough or inferred borough, ridership, fare/payment category                                 |
| <a href="#">Citi Bike trip data, April 2026</a>      | Compares short taxi trips with bike-share activity to assess Citi Bike alternative opportunities. | Started at, ended at, start station, end station, start/end coordinates, member type, rideable type                              |

Table 1: Data Sources and Analytical Purpose

### III. Data Engineering Pipeline

#### a) Data sourcing and Loading

The pipeline first set up a reproducible file management structure in Google Drive. Raw datasets were stored in a project folder with separate subfolders for taxi, lookup, subway, bus, bike and spatial files. This kept source files available after Colab runtime restarts and reduced repeated downloads of large datasets.

| Dataset                 | File type                  | Download method        | Stored path                                 | File size |
|-------------------------|----------------------------|------------------------|---------------------------------------------|-----------|
| Yellow taxi             | Parquet                    | Direct file download   | Google Drive /raw_data/taxi/                | 62.31 MB  |
| Green taxi              | Parquet                    | Direct file download   | Google Drive /raw_data/taxi/                | 1.03 MB   |
| Taxi zone lookup        | CSV                        | Direct file download   | Google Drive /raw_data/lookup/              | 0.01 MB   |
| Subway hourly ridership | CSV                        | Socrata API pagination | Google Drive /raw_data/subway/              | 288.59 MB |
| Bus hourly ridership    | CSV                        | Socrata API pagination | Google Drive /raw_data/bus/                 | 436.28 MB |
| Citi Bike               | ZIP containing 4 CSV files | Direct ZIP download    | Google Drive /raw_data/bike/                | 156.97 MB |
| Taxi zone shapefile     | ZIP/shapefile              | Direct ZIP download    | Google Drive /raw_data/taxi_zone_shapefile/ | N/A       |

*Table 2: Raw data file used in the pipeline*

The notebook used a download-if-missing approach, meaning each file was downloaded only if it was not already available in the expected Google Drive path. Direct links were used for the yellow and green taxi Parquet files, taxi zone lookup CSV, Citi Bike ZIP file and taxi zone shapefile ZIP. The larger MTA subway and bus datasets were downloaded through the Socrata API using pagination, selected fields and an April 2026 date filter.

The Citi Bike dataset required an extra extraction step because the April 2026 ZIP contained four CSV files. These were read with consistent data types, combined into one raw Citi Bike table, and saved as a separate CSV so later notebook runs could reuse the combined file without repeating the extraction and concatenation process.

Raw data was loaded using both DuckDB and pandas. The yellow and green taxi Parquet files were queried directly with DuckDB because it can scan large Parquet files efficiently without loading the full datasets into pandas memory. The taxi zone lookup, subway, bus and Citi Bike files were loaded with pandas. Schema and sample record checks for each raw dataset are provided in [Appendix A](#).

Yellow taxi sample:

|   | VendorID | tpep_pickup_datetime | tpep_dropoff_datetime | passenger_count | trip_distance | RatecodeID | store_and_fwd_flag | PULocationID |
|---|----------|----------------------|-----------------------|-----------------|---------------|------------|--------------------|--------------|
| 0 | 1        | 2026-04-01 00:40:05  | 2026-04-01 00:52:44   | 1               | 2.80          | 1          |                    | N            |
| 1 | 2        | 2026-04-01 00:09:19  | 2026-04-01 00:21:29   | 1               | 7.37          | 1          |                    | N            |
| 2 | 2        | 2026-04-01 00:15:29  | 2026-04-01 00:34:14   | 1               | 7.66          | 1          |                    | N            |
| 3 | 1        | 2026-04-01 00:14:20  | 2026-04-01 00:27:49   | 0               | 7.90          | 1          |                    | N            |
| 4 | 2        | 2026-04-01 00:04:53  | 2026-04-01 00:11:54   | 1               | 1.34          | 1          |                    | N            |

Table 3: Sample Raw Yellow Taxi Records

Yellow taxi schema:

|    | column_name           | column_type | null | key  | default | extra |
|----|-----------------------|-------------|------|------|---------|-------|
| 0  | VendorID              | INTEGER     | YES  | None | None    | None  |
| 1  | tpep_pickup_datetime  | TIMESTAMP   | YES  | None | None    | None  |
| 2  | tpep_dropoff_datetime | TIMESTAMP   | YES  | None | None    | None  |
| 3  | passenger_count       | BIGINT      | YES  | None | None    | None  |
| 4  | trip_distance         | DOUBLE      | YES  | None | None    | None  |
| 5  | RatecodeID            | BIGINT      | YES  | None | None    | None  |
| 6  | store_and_fwd_flag    | VARCHAR     | YES  | None | None    | None  |
| 7  | PULocationID          | INTEGER     | YES  | None | None    | None  |
| 8  | DOLocationID          | INTEGER     | YES  | None | None    | None  |
| 9  | payment_type          | BIGINT      | YES  | None | None    | None  |
| 10 | fare_amount           | DOUBLE      | YES  | None | None    | None  |
| 11 | extra                 | DOUBLE      | YES  | None | None    | None  |
| 12 | mta_tax               | DOUBLE      | YES  | None | None    | None  |
| 13 | tip_amount            | DOUBLE      | YES  | None | None    | None  |
| 14 | tolls_amount          | DOUBLE      | YES  | None | None    | None  |
| 15 | improvement_surcharge | DOUBLE      | YES  | None | None    | None  |
| 16 | total_amount          | DOUBLE      | YES  | None | None    | None  |
| 17 | congestion_surcharge  | DOUBLE      | YES  | None | None    | None  |
| 18 | Airport_fee           | DOUBLE      | YES  | None | None    | None  |
| 19 | cbd_congestion_fee    | DOUBLE      | YES  | None | None    | None  |

Table 4: Yellow Taxi Raw Data Schema

|   | VendorID | lpep_pickup_datetime | lpep_dropoff_datetime | store_and_fwd_flag | RatecodeID | PULocationID | DOLocationID | passenger |
|---|----------|----------------------|-----------------------|--------------------|------------|--------------|--------------|-----------|
| 0 | 2        | 2026-04-01 00:29:09  | 2026-04-01 00:54:48   | N                  | 1          | 66           | 48           |           |
| 1 | 2        | 2026-04-01 00:19:13  | 2026-04-01 00:19:20   | N                  | 5          | 129          | 129          |           |
| 2 | 2        | 2026-04-01 00:40:24  | 2026-04-01 00:47:31   | N                  | 1          | 74           | 141          |           |
| 3 | 2        | 2026-04-01 00:27:31  | 2026-04-01 00:35:33   | N                  | 1          | 244          | 116          |           |
| 4 | 2        | 2026-04-01 00:51:45  | 2026-04-01 00:58:33   | N                  | 1          | 243          | 235          |           |

Table 5: Sample Raw Green Taxi Records

|    | column_name           | column_type | null | key  | default | extra |
|----|-----------------------|-------------|------|------|---------|-------|
| 0  | VendorID              | INTEGER     | YES  | None | None    | None  |
| 1  | lpep_pickup_datetime  | TIMESTAMP   | YES  | None | None    | None  |
| 2  | lpep_dropoff_datetime | TIMESTAMP   | YES  | None | None    | None  |
| 3  | store_and_fwd_flag    | VARCHAR     | YES  | None | None    | None  |
| 4  | RatecodeID            | BIGINT      | YES  | None | None    | None  |
| 5  | PULocationID          | INTEGER     | YES  | None | None    | None  |
| 6  | DOLocationID          | INTEGER     | YES  | None | None    | None  |
| 7  | passenger_count       | BIGINT      | YES  | None | None    | None  |
| 8  | trip_distance         | DOUBLE      | YES  | None | None    | None  |
| 9  | fare_amount           | DOUBLE      | YES  | None | None    | None  |
| 10 | extra                 | DOUBLE      | YES  | None | None    | None  |
| 11 | mta_tax               | DOUBLE      | YES  | None | None    | None  |
| 12 | tip_amount            | DOUBLE      | YES  | None | None    | None  |
| 13 | tolls_amount          | DOUBLE      | YES  | None | None    | None  |
| 14 | ehail_fee             | DOUBLE      | YES  | None | None    | None  |
| 15 | improvement_surcharge | DOUBLE      | YES  | None | None    | None  |
| 16 | total_amount          | DOUBLE      | YES  | None | None    | None  |
| 17 | payment_type          | BIGINT      | YES  | None | None    | None  |
| 18 | trip_type             | BIGINT      | YES  | None | None    | None  |
| 19 | congestion_surcharge  | DOUBLE      | YES  | None | None    | None  |
| 20 | cbd_congestion_fee    | DOUBLE      | YES  | None | None    | None  |

Table 6: Green Taxi Raw Data Schema



|   | Dataset          | Rows loaded | Columns loaded | Role in project                     |
|---|------------------|-------------|----------------|-------------------------------------|
| 0 | Yellow taxi      | 3800664     | 20             | Anchor taxi demand dataset          |
| 1 | Green taxi       | 44238       | 21             | Additional taxi demand dataset      |
| 2 | Taxi zone lookup | 265         | 4              | Location ID to borough/zone lookup  |
| 3 | Subway ridership | 1724852     | 12             | Public transport comparison dataset |
| 4 | Bus ridership    | 7378560     | 6              | Public transport comparison dataset |
| 5 | Citi Bike        | 3860371     | 13             | Bike-share supporting context       |

Table 7: Dataset Loading Summary

After the initial inspection, all raw datasets were registered as DuckDB views: yellow\_taxi\_raw, green\_taxi\_raw, taxi\_zone\_raw, subway\_raw, bus\_raw and bike\_raw. This created a consistent raw data layer for the rest of the pipeline.

## b) Raw data audit

After the raw datasets were loaded and registered, the pipeline conducted an initial data audit before cleaning and transformation. Issue severity was classified using percentage-based thresholds to prioritise cleaning decisions (See [Appendix B](#) for classification rule)

| Dataset          | Audit rule                     | Justification                                                                                      |
|------------------|--------------------------------|----------------------------------------------------------------------------------------------------|
| Taxi             | April 2026 only                | Keeps all datasets within the same monthly scope for fair comparison.                              |
|                  | Drop-off after pickup          | Ensures each trip has a valid time order for duration and time-period analysis.                    |
|                  | Taxi distance > 0              | Removes zero or negative distances that cannot support distance analysis.                          |
|                  | Taxi distance ≤ 100 miles      | Keeps plausible long trips while removing extreme distance outliers.                               |
|                  | Fare/total amount ≥ 0          | Removes refunds, corrections or abnormal negative payment records.                                 |
|                  | Total amount ≤ \$500           | Limits extreme cost outliers that could distort average fare analysis.                             |
|                  | Location ID 1–263              | Keeps records that can be reliably joined to TLC taxi zones.                                       |
|                  | Passenger count audit          | Documents occupancy issues, but does not remove trips because the analysis focuses on trip demand. |
| Taxi zone lookup | Unique taxi zone LocationID    | Prevents duplicate or missing lookup keys from affecting geographic joins.                         |
| Subway           | Subway required fields         | Ensures records can support station, borough and time-period aggregation.                          |
|                  | Subway ridership/transfers ≥ 0 | Keeps count-based values logically valid for aggregation.                                          |

|           |                                         |                                                                                      |
|-----------|-----------------------------------------|--------------------------------------------------------------------------------------|
| Bus       | Bus required fields                     | Ensures records can support route, time-period and comparison analysis.              |
|           | Bus ridership/transfers $\geq 0$        | Keeps count-based values logically valid for aggregation.                            |
|           | Bus route-prefix review                 | Treats uncommon route groups carefully without automatically removing them.          |
| Citi Bike | Unique Citi Bike ride_id                | Prevents duplicate bike trips from being counted twice.                              |
|           | Citi Bike end time after start time     | Ensures valid ride duration calculation.                                             |
|           | Citi Bike duration $\leq 1,440$ minutes | Removes rides longer than 24 hours as abnormal for bike-share analysis.              |
|           | Missing Citi Bike location fields       | Keeps records for hourly demand but excludes/flags them for station or map analysis. |

Table 3: Data Audit Rules and Justifications

The taxi datasets required the most detailed audit because they were the anchor source for the analysis. The raw taxi data contained 3,800,664 yellow taxi records and 44,238 green taxi records. The date range check showed that most records were within the April 2026 scope, but some taxi records fell outside the study period, including a small number of yellow taxi records with historical or May timestamps and green taxi records extending into early May. This justified filtering the cleaned taxi table to April 2026 pickup dates only.

| index | taxi_type | total_rows | min_pickup_datetime | max_pickup_datetime | min_dropoff_datetime | max_dropoff_datetime |
|-------|-----------|------------|---------------------|---------------------|----------------------|----------------------|
| 0     | green     | 44238      | 2026-03-31 23:28:50 | 2026-05-01 07:53:18 | 2026-04-01 00:15:53  | 2026-05-02 07:07:21  |
| 1     | yellow    | 3800664    | 2001-01-01 09:23:58 | 2026-05-01 00:01:28 | 2001-01-01 16:09:38  | 2026-05-02 12:06:09  |

Table 4: Taxi Date Coverage Check Before Cleaning

| Metric                       | Value  | Metric                       | Value     |
|------------------------------|--------|------------------------------|-----------|
| Taxi Type                    | Green  | Taxi Type                    | Yellow    |
| Total Rows                   | 44,238 | Total Rows                   | 3,800,664 |
| Missing Pickup Datetime      | 0      | Missing Pickup Datetime      | 0         |
| Missing Dropoff Datetime     | 0      | Missing Dropoff Datetime     | 0         |
| Invalid Time Order           | 38     | Invalid Time Order           | 18935     |
| Missing Trip Distance        | 0      | Missing Trip Distance        | 0         |
| Non-Positive Trip Distance   | 1,604  | Non-Positive Trip Distance   | 93,150    |
| Extreme Trip Distance        | 9      | Extreme Trip Distance        | 136       |
| Missing Total Amount         | 0      | Missing Total Amount         | 0         |
| Negative Total Amount        | 155    | Negative Total Amount        | 14874     |
| Extreme Total Amount         | 5      | Extreme Total Amount         | 82        |
| Missing Fare Amount          | 0      | Missing Fare Amount          | 0         |
| Negative Fare Amount         | 153    | Negative Fare Amount         | 14328     |
| Missing Passenger Count      | 6,290  | Missing Passenger Count      | 799,786   |
| Non-Positive Passenger Count | 543    | Non-Positive Passenger Count | 11692     |
| Unusual Passenger Count      | 9      | Unusual Passenger Count      | 4         |
| Missing Pickup Location      | 0      | Missing Pickup Location      | 0         |
| Missing Dropoff Location     | 0      | Missing Dropoff Location     | 0         |
| Invalid Pickup Location ID   | 139    | Invalid Pickup Location ID   | 6013      |
| Invalid Dropoff Location ID  | 733    | Invalid Dropoff Location ID  | 20383     |

Table 5: Green & Yellow Taxi Raw Data Quality Audit

The taxi quality audit identified issues that could affect time, distance, cost and location-based analysis. Yellow taxis had 112,221 records with invalid time or distance values, while green taxis had 1,651. These included invalid time order, non-positive distances and extreme distances. Fare issues were also found, with 29,284 invalid or extreme yellow taxi fare records and 313 green taxi records.

Location fields were checked because taxi-zone analysis depends on valid pickup and drop-off IDs. The audit found 26,396 yellow taxi records and 872 green taxi records with invalid location IDs. No duplicate taxi trip records were identified. Passenger count issues were documented as a data quality limitation but not used as a main exclusion rule because the project focuses on completed taxi trips rather than occupancy.

| index | taxi_type | issue                                            | affected_rows | affected_pct | severity | recommended_action                                                        |
|-------|-----------|--------------------------------------------------|---------------|--------------|----------|---------------------------------------------------------------------------|
| 0     | green     | Missing or unusual passenger count               | 6,842         | 15.47%       | High     | Keep for trip-demand analysis, but exclude from passenger-based metrics   |
| 1     | green     | Invalid trip time or distance values             | 1,651         | 3.73%        | Medium   | Remove invalid time-order rows and non-positive or extreme distance trips |
| 2     | green     | Date coverage outside expected April 2026 period | N/A           | N/A          | Medium   | Filter records to April 2026 before analysis                              |
| 3     | green     | Invalid pickup or dropoff location IDs           | 872           | 1.97%        | Low      | Exclude invalid location IDs from borough and zone-level analysis         |
| 4     | green     | Invalid or extreme fare values                   | 313           | 0.71%        | Low      | Remove negative fare/payment rows and review extreme fare outliers        |
| 5     | green     | Potential duplicate trip records                 | 0             | 0.00%        | None     | No duplicate issue identified                                             |
| 6     | yellow    | Missing or unusual passenger count               | 811,482       | 21.35%       | High     | Keep for trip-demand analysis, but exclude from passenger-based metrics   |
| 7     | yellow    | Invalid trip time or distance values             | 112,221       | 2.95%        | Medium   | Remove invalid time-order rows and non-positive or extreme distance trips |
| 8     | yellow    | Date coverage outside expected April 2026 period | N/A           | N/A          | Medium   | Filter records to April 2026 before analysis                              |
| 9     | yellow    | Invalid pickup or dropoff location IDs           | 26,396        | 0.69%        | Low      | Exclude invalid location IDs from borough and zone-level analysis         |
| 10    | yellow    | Invalid or extreme fare values                   | 29,284        | 0.77%        | Low      | Remove negative fare/payment rows and review extreme fare outliers        |
| 11    | yellow    | Potential duplicate trip records                 | 0             | 0.00%        | None     | No duplicate issue identified                                             |

*Table 6: Taxi Data Quality Issues, Severity and Recommended Actions*

The taxi zone lookup table was suitable for geographic enrichment, with 265 rows, 265 unique LocationID values, no duplicate IDs, and no missing borough, zone or service-zone values. However, three special borough categories, N/A, EWR and Unknown, were identified and retained carefully because they may appear in taxi joins but should not be treated as standard NYC boroughs in borough-level analysis (See [Appendix C](#)).

The subway dataset was the cleanest supporting dataset, with 1,724,852 hourly records from 1–30 April 2026. It had no missing key fields, no negative ridership or transfer values, and covered 428 unique station complexes across five boroughs.

Therefore, it only required light cleaning, mainly timestamp conversion and time period feature creation (See [Appendix C](#)).

The bus dataset also showed strong completeness, with 7,378,560 hourly records, 427 unique routes, no missing key fields, and no negative ridership or transfer values. However, because bus data is route-based rather than taxi-zone based, route prefixes such as Q, B, M, BX,...were used only to create an approximate borough proxy for high-level comparison (See [Appendix C](#)).

The Citi Bike dataset contained 3,860,371 records and no duplicate ride IDs. It covered April 2026, with no missing start/end timestamps and no invalid time order records. The main issues were minor location gaps, including missing station or coordinate fields. These records were kept for overall demand analysis but flagged or excluded where station-level or map-based analysis required complete location data (See [Appendix C](#)).

Overall, the audit showed that taxi data required the most cleaning, while subway, bus and Citi Bike data were mostly suitable for supporting comparison. These audit findings informed the cleaning rules in Table 12.

| index | Dataset          | Key audit findings                                                                                                                                                                                                                                                                                                                                                                  | Severity       | Cleaning implication                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | Taxi             | Main issues were missing or unusual passenger counts (21.35% of yellow trips; 15.47% of green trips), invalid time or distance values (2.95% of yellow trips; 3.73% of green trips), invalid pickup/drop-off location IDs (0.69% of yellow trips; 1.97% of green trips), and a small number of invalid or extreme fare values. No potential duplicate trip records were identified. | Medium to High | Filter taxi records to valid April 2026 trips, remove invalid time-order rows, remove non-positive or extreme trip distances, remove negative or extreme fare/payment values, and keep only valid pickup/drop-off location IDs for borough and zone-level analysis. Do not remove trips only because passenger count is missing or unusual, because the main analysis focuses on trip demand rather than passenger occupancy. |
| 1     | Taxi zone lookup | No duplicate or missing key zone fields were identified. However, 3 records (1.13%) belong to special borough categories such as N/A, EWR, or Unknown.                                                                                                                                                                                                                              | Low to Medium  | Use the lookup table for taxi location enrichment. Keep valid TLC special categories in the raw lookup, but exclude or report N/A, EWR, and Unknown separately in NYC borough-level analysis.                                                                                                                                                                                                                                 |
| 2     | Subway           | No missing values were found in key fields. No negative ridership or transfer values were identified. The dataset covers April 2026, 5 boroughs, and 428 station complexes.                                                                                                                                                                                                         | Low            | Apply light cleaning only: cast timestamps, ridership, and transfer fields to appropriate data types; create time-period features; aggregate to borough, station, and time-period levels for comparison with taxi demand.                                                                                                                                                                                                     |
| 3     | Bus              | No missing key fields or negative ridership/transfer values were identified. However, 155,520 records (2.11%) relate to very low or zero ridership route prefixes, which should be treated carefully in comparison analysis.                                                                                                                                                        | Low to Medium  | Apply light cleaning only: cast timestamps, ridership, and transfer fields to appropriate data types; create time-period and route-prefix features. Treat route-prefix borough mapping as an approximate proxy, not an exact spatial match.                                                                                                                                                                                   |
| 4     | Citi Bike        | No duplicate ride IDs or invalid trip timestamps were identified. The dataset has complete April 2026 date coverage. However, 11,825 records (0.31%) have missing station information and 12,412 records (0.32%) have missing coordinate information.                                                                                                                               | Low            | Cast trip timestamps, calculate ride duration, and create time-period features. Keep records with missing station or coordinate values for overall bike demand analysis, but exclude them from station-level or coordinate-based analysis.                                                                                                                                                                                    |

Table 7: Dataset-Level Audit Findings and Cleaning Implications

### c) Cleaning and transformation

After the initial audit, the pipeline cleaned and transformed the raw datasets into analysis-ready DuckDB tables. The cleaning rules were based on the audit findings and were designed to protect the reliability of the business analysis.

| index | Dataset          | Audit finding                                                                                    | Cleaning / transformation decision                                                                                                      |
|-------|------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 0     | Taxi             | Some records fall outside April 2026 or have invalid pickup/drop-off timestamps                  | Keep only trips with pickup dates in April 2026 and drop-off after pickup                                                               |
| 1     | Taxi             | Some records have invalid time order, non-positive distance, or extreme distance values          | Keep trips with positive distance and remove extreme distance outliers above 100 miles                                                  |
| 2     | Taxi             | Some records have negative or extreme fare/payment values                                        | Keep records with non-negative fare and total amount values; remove extreme total amounts above \$500                                   |
| 3     | Taxi             | Some records have invalid pickup or drop-off location IDs                                        | Keep records with pickup and drop-off location IDs between 1 and 263                                                                    |
| 4     | Taxi zone lookup | Lookup is mostly clean but contains special borough categories such as N/A, EWR, and Unknown     | Use lookup for enrichment; keep special categories but flag whether pickup borough is one of the five NYC boroughs                      |
| 5     | Subway           | No major missing or negative-value issues identified                                             | Cast fields to correct data types and create time-period features                                                                       |
| 6     | Bus              | No major missing or negative-value issues identified, but route-prefix comparison is approximate | Cast fields to correct data types, create time-period and route-prefix features, and treat borough proxy cautiously                     |
| 7     | Citi Bike        | Mostly clean, but some records have missing station or coordinate information                    | Cast timestamps, calculate ride duration, keep valid trip times, and retain missing station/coordinate rows for overall demand analysis |

*Table 8: Final Cleaning and Transformation Decisions*

Yellow and green taxi records were standardised into a common structure and combined into taxi\_clean. The cleaning filters retained only April 2026 pickup records, required drop-off time to occur after pickup time, kept positive trip distances up to 100 miles, removed negative fare and total amount values, removed total amounts above \$500, and kept only pickup and drop-off location IDs within the valid taxi zone range. These rules were applied because invalid timestamps would distort hourly and time-period analysis, invalid distances would distort short- and long-trip patterns, abnormal fares would distort cost analysis, and invalid location IDs would prevent reliable zone-level mapping.

| index | taxi_type | cleaned_rows | min_pickup_datetime | max_pickup_datetime | avg_trip_distance | avg_total_amount |
|-------|-----------|--------------|---------------------|---------------------|-------------------|------------------|
| 0     | green     | 42090        | 2026-04-01 00:05:24 | 2026-04-30 23:54:33 | 3.33              | 25.09            |
| 1     | yellow    | 3653911      | 2026-04-01 00:00:00 | 2026-04-30 23:59:59 | 3.46              | 29.73            |

*Table 9: Cleaned Taxi Dataset Summary*

Outlier thresholds were applied conservatively. The 100-mile distance threshold retained plausible long taxi trips, such as airport or cross-borough travel, while removing extreme values that could distort average distance and long-trip metrics. Similarly, the \$500 total amount threshold kept expensive but realistic trips while reducing the influence of extreme payment values.

The taxi\_enriched table adds borough and zone information to the cleaned taxi trips by joining taxi location IDs with the taxi zone lookup table. This makes the data suitable for geographic analysis.



| Metric                  | Value     |
|-------------------------|-----------|
| Total Rows              | 3,696,001 |
| Missing Pickup Borough  | 0         |
| Missing Dropoff Borough | 0         |
| Unique Pickup Boroughs  | 6         |
| Unique Pickup Zones     | 258       |
| NYC Pickup Rows         | 3,695,902 |

Table 10: Taxi Zone Enrichment Validation Summary

The next transformation created taxi\_features. Pickup timestamps were converted into analytical time variables, including pickup hour, day of week, day name, weekend flag and time period (refer to [Appendix D](#)). Time periods were grouped into morning peak, evening peak, late night and off-peak so that taxi demand could be compared across meaningful planning periods. Trip distance was classified into short, medium and long trip categories. These features directly supported the business question by separating overall taxi volume from specific mobility signals such as late-night reliance, short-trip concentration and long-distance travel.

| Feature      | Rule                          | Justification                                                 |
|--------------|-------------------------------|---------------------------------------------------------------|
| pickup_hour  | Extract hour from pickup time | Supports hourly demand analysis.                              |
| is_weekend   | Saturday/Sunday = 1           | Separates weekend from weekday travel patterns.               |
| Morning peak | 7:00–9:59                     | Captures typical morning commute activity.                    |
| Evening peak | 16:00–19:59                   | Captures afternoon/evening travel and pickup pressure.        |
| Late night   | 22:00–5:59                    | Captures taxi reliance outside standard daytime/peak periods. |
| Off peak     | All other hours               | Groups remaining non-peak demand.                             |
| Short trip   | < 2 miles                     | Supports bike/walking/bus alternative investigation.          |
| Medium trip  | 2–6 miles                     | Represents moderate urban trips.                              |
| Long trip    | > 6 miles                     | Supports connectivity and airport/cross-borough review.       |

Table 11: Engineered Feature Definitions and Justifications

The supporting public and active transport datasets required lighter cleaning because the audit found fewer validity issues (refer to [Appendix D](#)). Subway records were transformed into subway\_clean by standardising timestamp, ridership, transfer



and location fields, keeping April 2026 records, and confirming that ridership and transfers were non-negative. Bus records were transformed into bus\_clean using similar timestamp and ridership checks. A route-prefix feature was also created for broad borough-level comparison because the bus dataset does not provide taxi-zone-level geography. This proxy followed MTA route-naming conventions but was used only as an approximate comparison, not as a precise geographic join or evidence of exact bus service gaps (MTA, 2026).

Citi Bike records were transformed into bike\_clean by standardising ride IDs, bike type, timestamps, station fields, coordinates and user type (refer to Appendix D for results). Ride duration was calculated in minutes, and only records with valid time order and duration of 1,440 minutes or less were retained. Missing station or coordinate fields were kept for overall hourly bike-demand analysis because they still supported time-based comparison, but were flagged or excluded from station-level and map-based analysis.

The cleaning stage was validated using before and after row counts and table-level checks. The combined taxi dataset was reduced from 3,844,902 raw records to 3,696,001 cleaned records, removing 148,901 records or 3.87%. Citi Bike was reduced from 3,860,371 to 3,859,605 records, removing only 766 records or 0.02%. Subway and bus records were retained after standardisation because no invalid timestamps or negative ridership and transfer values required removal.

|   | dataset       | rows_before_cleaning | rows_after_cleaning | rows_removed | pct_removed |
|---|---------------|----------------------|---------------------|--------------|-------------|
| 0 | Bus           | 7378560              | 7378560             | 0            | 0.00%       |
| 1 | Citi Bike     | 3860371              | 3859605             | 766          | 0.02%       |
| 2 | Subway        | 1724852              | 1724852             | 0            | 0.00%       |
| 3 | Taxi combined | 3844902              | 3696001             | 148,901      | 3.87%       |

Table 12: Before-and-After Cleaning Row Counts

Overall, the cleaning and transformation process produced six main analysis-ready tables: taxi\_clean, taxi\_enriched, taxi\_features, subway\_clean, bus\_clean and bike\_clean.

#### d) SQL analysis and engineered summary tables

Instead of analysing raw records directly, the pipeline stored engineered outputs as DuckDB tables so that the same results could be reused consistently in the visualisation, data story and recommendation sections.

The SQL analysis was organised into four layers: taxi demand pattern analysis, supporting mode summaries, mode-comparison tables, and intervention priority grouping (Refer to [Appendix E](#))

| SQL layer                      | Tables created                                                                                                                                                                                                                              | Purpose                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Taxi demand pattern analysis   | taxi_borough_summary,<br>taxi_hourly_summary,<br>taxi_borough_time_period_summary,<br>taxi_borough_distance_summary,<br>taxi_zone_summary,<br>taxi_late_night_zone_summary,<br>taxi_short_trip_zone_summary,<br>taxi_long_trip_zone_summary | Summarises where and when taxi demand is highest, and identifies late-night, short-trip and long-trip patterns by borough and zone. |
| Supporting mode summaries      | subway_borough_time_summary,<br>bus_borough_proxy_time_summary,<br>bike_time_period_summary                                                                                                                                                 | Aggregates subway, bus and Citi Bike activity into comparable time-period summaries for supporting context.                         |
| Mode-comparison on tables      | taxi_subway_comparison,<br>taxi_bus_comparison,<br>taxi_bike_hourly_comparison                                                                                                                                                              | Compares taxi activity with subway, bus and Citi Bike patterns to support cautious interpretation of taxi reliance.                 |
| Intervention priority grouping | taxi_zone_intervention_priority,<br>intervention_group_summary                                                                                                                                                                              | Converts taxi-zone indicators into recommended intervention categories and summary counts.                                          |

*Table 13: Engineered SQL tables created for analysis*

The taxi demand pattern tables formed the core of the SQL analysis because the business question focused on taxi zone intervention priorities. Borough, hourly and time-period summaries were used to identify where taxi demand was concentrated and how it varied by time of day, trip distance, cost, duration, late-night share, short-trip share and long-trip share. Distance and zone-level summaries then provided more detailed evidence on whether each area showed short-trip pressure, late-night reliance or longer distance connectivity needs.

The main zone-level table, taxi\_zone\_summary, summarised taxi trips, average distance, average cost, and late-night, short-trip and long-trip shares for each pickup zone. Three supporting zone tables, taxi\_late\_night\_zone\_summary, taxi\_short\_trip\_zone\_summary and taxi\_long\_trip\_zone\_summary, were created to identify zones with specific demand signals. This ensured the recommendation logic considered the type of taxi demand, not only total trip volume.

Supporting mode summaries placed taxi demand in a wider mobility context. Subway, bus and Citi Bike tables aggregated ridership or activity by borough, time period or hour. The bus comparison used a route prefix borough proxy, so it was interpreted cautiously as an approximate comparison only. Mode comparison tables then compared taxi activity with subway, bus and Citi Bike patterns to support issue identification, without claiming that taxis directly replaced other modes.

The final SQL output, taxi\_zone\_intervention\_priority, classified taxi zones using four signals: high-taxi-volume, late-night concentration, short-trip concentration and long-trip concentration. Thresholds were based on percentile cut-offs from taxi\_zone\_summary, using the 80<sup>th</sup> percentile for taxi volume and the 75<sup>th</sup> percentile for the three demand-share measures. Minimum trip-count filters were applied to avoid prioritising zones based on high percentages from small samples. A priority score

was then calculated from the number of signals present, and each zone was assigned a recommended intervention category.

| Classification rule                                             | Why it was used                                                                             |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| total_taxi_trips >= 1000                                        | Removes very low-volume zones so priority signals are based on meaningful activity.         |
| 80th percentile for high taxi volume                            | Identifies the highest-demand taxi zones relative to the April 2026 dataset.                |
| 75th percentile for late-night, short-trip and long-trip shares | Identifies zones with unusually strong trip-pattern signals compared with other zones.      |
| Late-night share + at least 500 late-night trips                | Prevents high percentages from small late-night trip counts being over-prioritised.         |
| Short-trip share + at least 1,000 short trips                   | Ensures short-trip alternative review is based on both concentration and sufficient volume. |
| Long-trip share + at least 1,000 long trips + high cost         | Targets zones where long-distance taxi use is both common and financially significant.      |
| Weighted priority score                                         | Combines multiple signals while giving more importance to high-volume operational pressure. |
| Intervention hierarchy                                          | Keeps the recommendation map simple and assigns each zone one main intervention category.   |

Table 14: Mobility Intervention Classification Rules

The categories included taxi pickup/drop-off management, late-night public transport support review, short-trip alternative review, subway or public transport connectivity review and lower priority monitoring.

| index | pickup_borough | pickup_zone               | total_taxi_trips | priority_level | recommended_intervention        | priority_score | signal_count | signal_summary                                                                             | late_night_trip_share_pct | short_trip_share_pct |
|-------|----------------|---------------------------|------------------|----------------|---------------------------------|----------------|--------------|--------------------------------------------------------------------------------------------|---------------------------|----------------------|
| 0     | Manhattan      | Times Sq/Theatre District | 109175           | High priority  | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi concentration; High short-trip concentration | 23.61                     | 58.43                |
| 1     | Manhattan      | East Village              | 93298            | High priority  | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi concentration; High short-trip concentration | 45.62                     | 53.45                |
| 2     | Manhattan      | Clinton East              | 83191            | High priority  | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi concentration; High short-trip concentration | 29.36                     | 58.19                |
| 3     | Manhattan      | West Village              | 80404            | High priority  | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi concentration; High short-trip concentration | 41.29                     | 60.41                |
| 4     | Manhattan      | Flatiron                  | 53384            | High priority  | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi concentration; High short-trip concentration | 23.89                     | 65.8                 |
| 5     | Manhattan      | Greenwich Village South   | 53280            | High priority  | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi concentration; High short-trip concentration | 47.3                      | 57.89                |

Table 15: Taxi Zone Intervention Priority

| index | recommended_intervention                      | zone_count | total_taxi_trips | avg_priority_score | avg_signal_count | avg_trip_distance_miles | avg_total_amount_usd | avg_late_night_share_pct | avg_short_trip_s |
|-------|-----------------------------------------------|------------|------------------|--------------------|------------------|-------------------------|----------------------|--------------------------|------------------|
| 0     | Taxi pickup/drop-off management               | 29         | 2651599.0        | 3.83               | 2.24             | 3.03                    | 28.53                | 18.59                    |                  |
| 1     | Late-night public transport support review    | 27         | 301805.0         | 1.65               | 1.11             | 4.59                    | 33.89                | 35.96                    |                  |
| 2     | Short-trip alternative review                 | 7          | 267956.0         | 1.5                | 1.0              | 2.57                    | 25.36                | 13.12                    |                  |
| 3     | Subway / public transport connectivity review | 10         | 24169.0          | 1.0                | 1.0              | 10.76                   | 47.36                | 13.97                    |                  |
| 4     | Lower priority / monitor                      | 69         | 404958.0         | 0.0                | 0.0              | 6.12                    | 35.07                | 15.01                    |                  |

Table 16: Intervention Category Summary

## IV. Data Governance, Privacy and Ethics

Although the datasets are public, they still need careful handling because detailed trip times and locations could reveal sensitive travel patterns.

| Governance risk                                                                 | Pipeline control applied                                                                                              |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Trip-level location and timestamp data could reveal sensitive movement patterns | Results were aggregated to hourly, borough, taxi-zone, route-proxy and intervention-category levels                   |
| Individual passenger or driver inference                                        | No individual trip paths, passenger identities, driver behaviour or repeated journeys were visualised                 |
| Unnecessary data use                                                            | The analysis focused on fields needed for time, location, distance, cost and demand analysis                          |
| Misleading public transport comparison                                          | Taxi, subway, bus and Citi Bike comparisons were interpreted as contextual evidence, not proof of direct substitution |
| Approximate bus geography                                                       | The bus borough proxy was clearly labelled as approximate and used only for broad comparison                          |
| Unsupported policy claims                                                       | Recommendations were framed as candidate areas for further planning review, not final infrastructure decisions        |

*Table 17: Governance Risks and Pipeline Controls*

The first control was data minimisation. The pipeline documented which fields were used and why, focusing on variables directly linked to the business question, such as trip time, location, borough, zone, distance, fare and total amount. Passenger count, vendor and payment fields, were not included (See [Appendix F](#)).

The second control was privacy-aware aggregation. Although the pipeline processed trip-level records, the report only presents aggregated results by borough, pickup zone, hour, time period and intervention group. Subway, bus and Citi Bike data were also summarised at broader time-period, borough-proxy or hourly levels. The report does not show individual travel paths or exact person level movements, reducing privacy risk.

The third control was cautious spatial and modal comparison. The datasets use different geographic units, including taxi zones, subway stations, bus routes and Citi Bike stations or coordinates. Therefore, taxi to subway, taxi to bus and taxi to bike comparisons were treated as screening indicators, not causal evidence. The bus comparison was especially cautious because its borough variable was based on route prefixes rather than exact boarding locations, so it was used only as broad context ([Appendix F](#)).

The fourth control was responsible wording. The report avoided overclaiming from aggregate data: high taxi volume was described as recorded pickup activity, late-night activity as a signal for service review, and short taxi trips as candidates for further alternative-mode investigation. (See [Appendix F](#)).

### \* Assumptions and Limitations

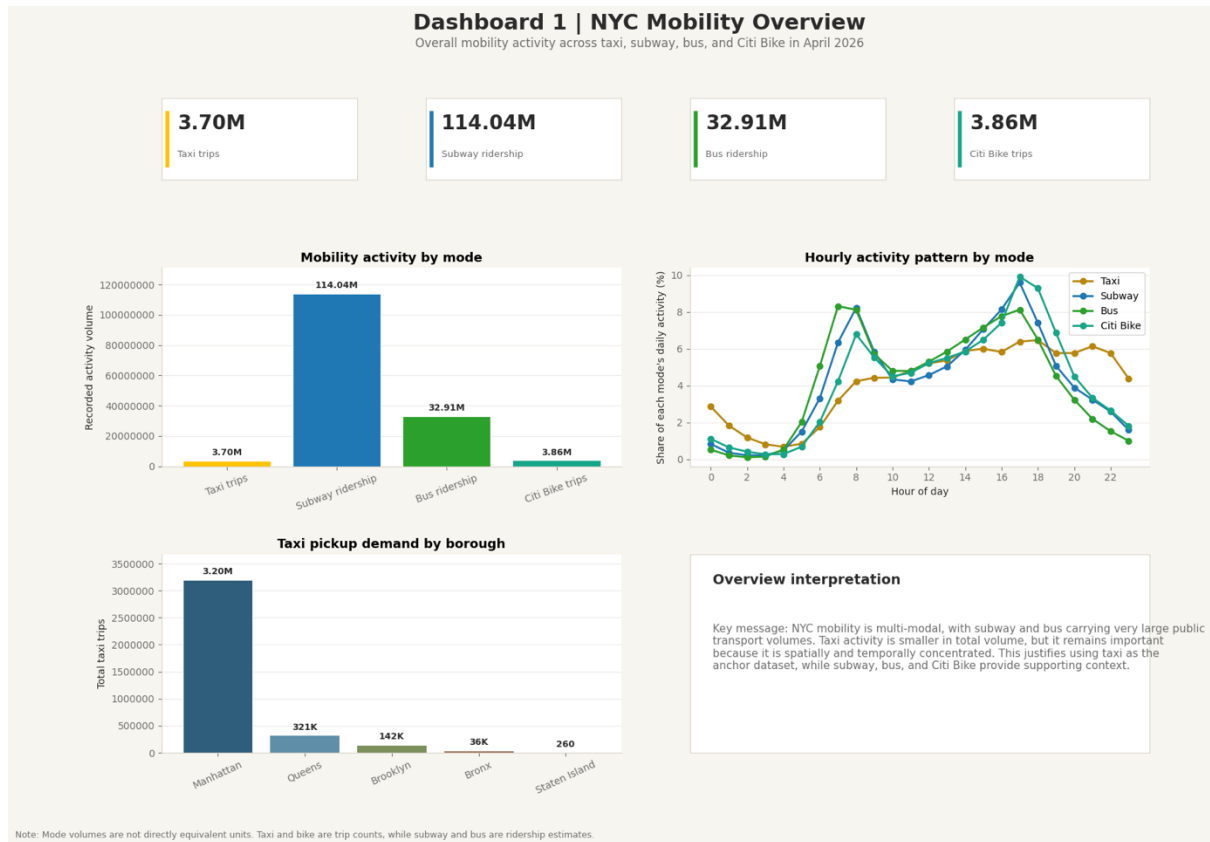
Several assumptions were applied to make the NYC mobility datasets comparable and suitable for analysis, however, they raises some limitations.

| Assumption                                                                                  | How it was applied                                                                                                  | Implication                                                                                                                                      |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| April 2026 is a suitable analysis period                                                    | All datasets were filtered or selected for April 2026.                                                              | Findings represent one month only and may not reflect seasonal or long-term patterns.                                                            |
| Completed taxi trips can be used as a proxy for taxi demand                                 | Yellow and green taxi trip counts were used to measure recorded taxi pickup activity.                               | The analysis captures completed taxi trips, not unmet demand or trips that could not access taxis.                                               |
| Pickup location represents the origin of taxi demand                                        | Taxi-zone analysis was mainly based on pickup borough and pickup zone.                                              | Recommendations focus on where taxi demand starts, not necessarily where passengers travel to.                                                   |
| Taxi zone lookup provides appropriate geographic interpretation                             | Pickup and drop-off location IDs were joined to borough and zone names.                                             | Taxi zones approximate neighbourhood-level geography but are not exact neighbourhood boundaries.                                                 |
| Yellow and green taxi data can be combined for taxi-demand analysis                         | Both datasets were cleaned into a common structure and analysed together.                                           | This gives a broader taxi view, but yellow and green taxis serve partly different markets.                                                       |
| Passenger count is not the main demand measure                                              | Passenger count was audited but not used as a core filtering or demand variable.                                    | The analysis measures taxi trip demand, not total passenger movement or vehicle occupancy.                                                       |
| Time-period categories are suitable planning bands                                          | Trips were grouped into morning peak, evening peak, late night and off peak.                                        | These categories simplify hourly patterns into planning-relevant periods, but they do not capture all travel-purpose differences.                |
| Short, medium and long taxi trip categories can indicate different mobility signals         | Trips under 2 miles were treated as short, 2–6 miles as medium, and above 6 miles as long.                          | These categories support intervention logic, but they do not prove that passengers should use another mode.                                      |
| Percentile thresholds can identify relatively high-priority zones                           | High-volume and high-share signals were based on percentile thresholds within April 2026 data.                      | Priority categories are relative to this dataset and may change if a different month or threshold is used.                                       |
| Minimum trip-count filters reduce misleading percentage effects                             | Zones needed sufficient trip volumes before being classified as having late-night, short-trip or long-trip signals. | This reduces the risk of over-prioritising zones with high percentages but very small trip counts.                                               |
| Subway, bus and Citi Bike data provide supporting context, not direct substitution evidence | These datasets were compared with taxi patterns by time period, borough proxy or hour.                              | Comparisons indicate possible areas for review, but they do not prove that taxis replace other modes.                                            |
| Bus route prefixes can approximate broad borough patterns                                   | Bus routes were grouped using route prefixes to create a borough proxy.                                             | This is suitable only for broad comparison, not precise geographic joins or exact bus service-gap claims.                                        |
| Citi Bike activity can support short-trip alternative analysis                              | Citi Bike trips were compared with short taxi trips by hour.                                                        | This identifies possible alternative-mode opportunities, but cycling suitability depends on station coverage, safety, weather and accessibility. |
| Aggregated outputs are appropriate for privacy-aware reporting                              | Findings were reported at borough, zone, route-proxy, time-period or intervention-group level.                      | This reduces privacy risk but limits individual-level interpretation.                                                                            |

*Table 18: Key Assumptions and Analytical Limitations*

## V. Data Story: Analysis and Findings

### a) Overview: taxi demand is concentrated by place and time



Overall, taxi demand in April 2026 was highly concentrated geographically. Manhattan dominated the cleaned taxi dataset with 3.20 million pickup trips, followed by Queens with about 321,000 trips and Brooklyn with about 142,000 trips. This indicates that taxi activity was unevenly distributed across New York City.



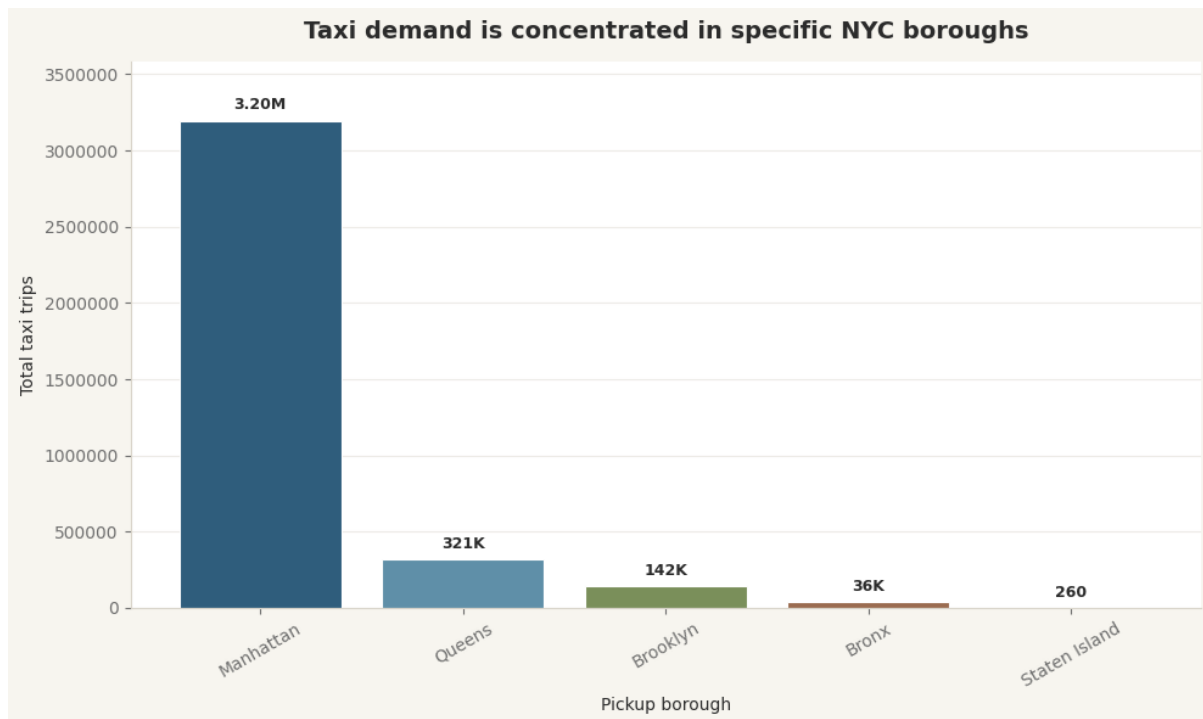


Figure 1: Taxi Demand by Pickup Borough

This finding matters because high taxi volume creates a different planning issue from lower-volume but late-night or long-distance taxi use. In Manhattan, the main concern is likely to be operational pressure around curb space, pickup/drop-off points and congestion in high-activity areas. Therefore, high-volume Manhattan zones are more suitable for taxi pickup/drop-off management than for broad claims about public transport gaps. The full borough-level taxi summary is provided in [Appendix G](#).

The zone-level analysis provides a more actionable view. As shown in Figure 2, the highest-volume pickup zones included dense Manhattan activity centres as well as major airport zones such as JFK Airport and LaGuardia Airport. These two types of high demand zones suggest different planning issues. Dense Manhattan zones point to curbside and pickup/drop-off management, while airport zones also indicate airport access and longer-distance travel patterns. The complete pickup-zone ranking are provided in [Appendix G](#).

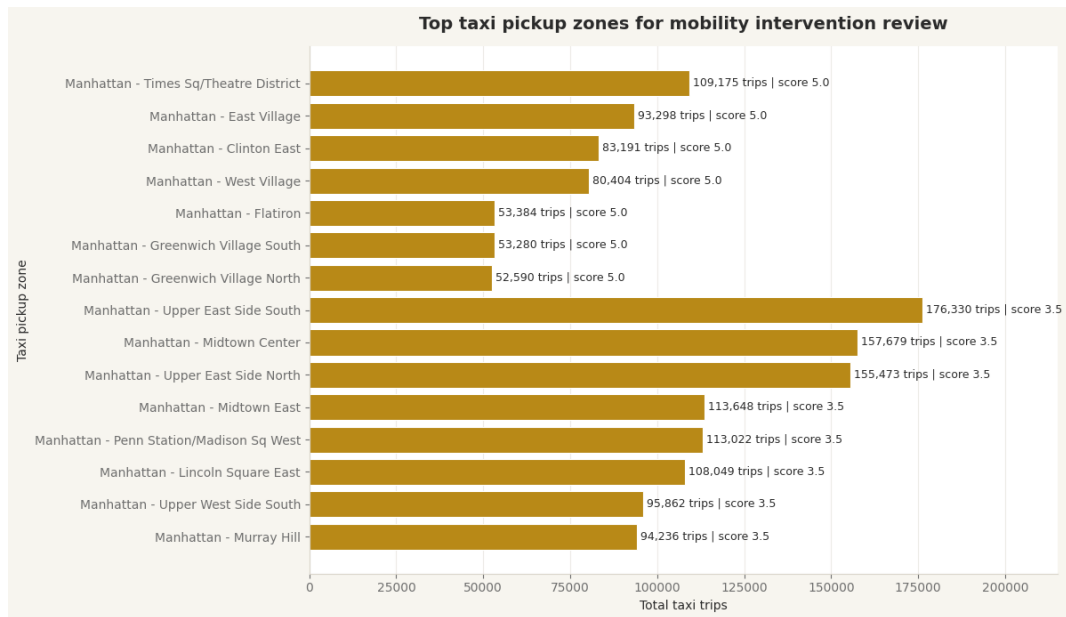


Figure 2: Top Taxi Pickup Zones for Mobility Intervention Review

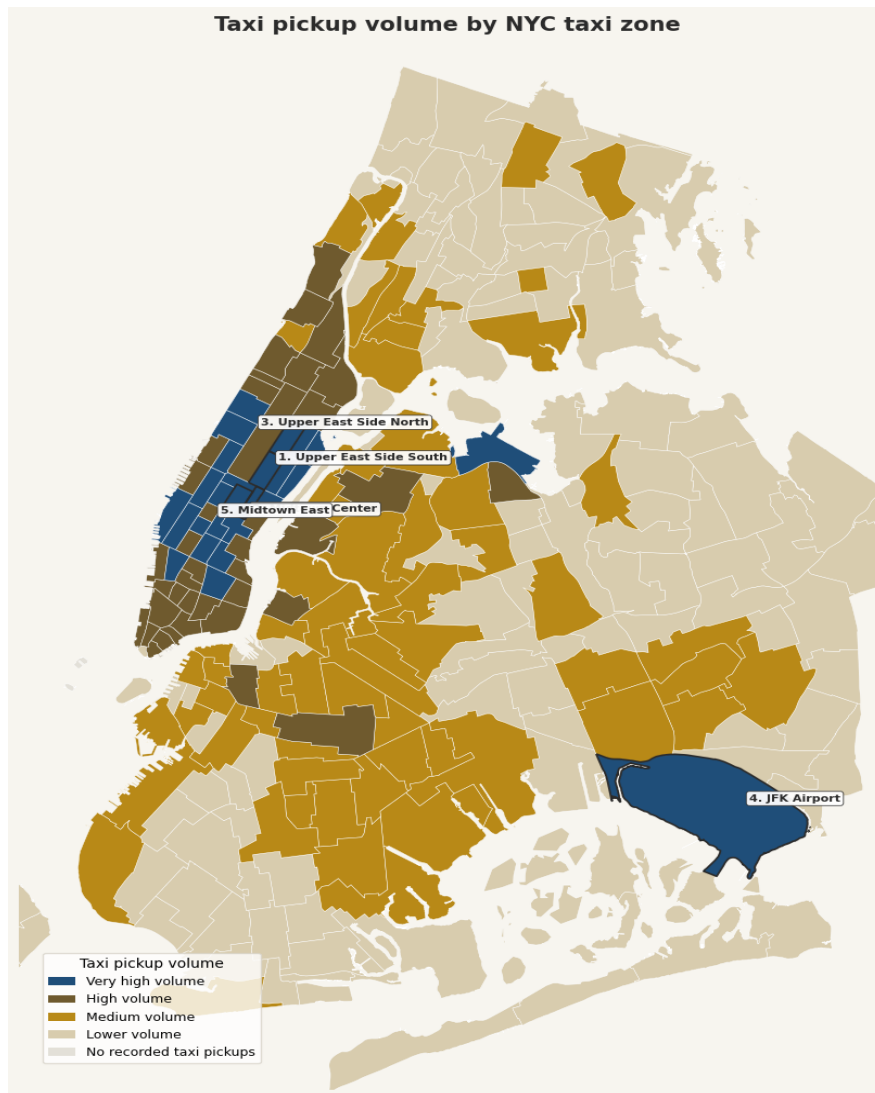


Figure 3: Taxi Pickup Volume by NYC Taxi Zone Map

The hourly results show that taxi demand was strongest later in the day. The highest hourly volumes occurred at 18:00, 17:00 and 21:00, while demand was lowest in the early morning around 04:00. This indicates that taxi planning should not focus only on the morning commute. Evening and late-night activity should also be considered because taxi demand may increase when work, leisure, dining, events and night-time travel overlap. The full taxi hourly demand summary is provided in [Appendix G](#).

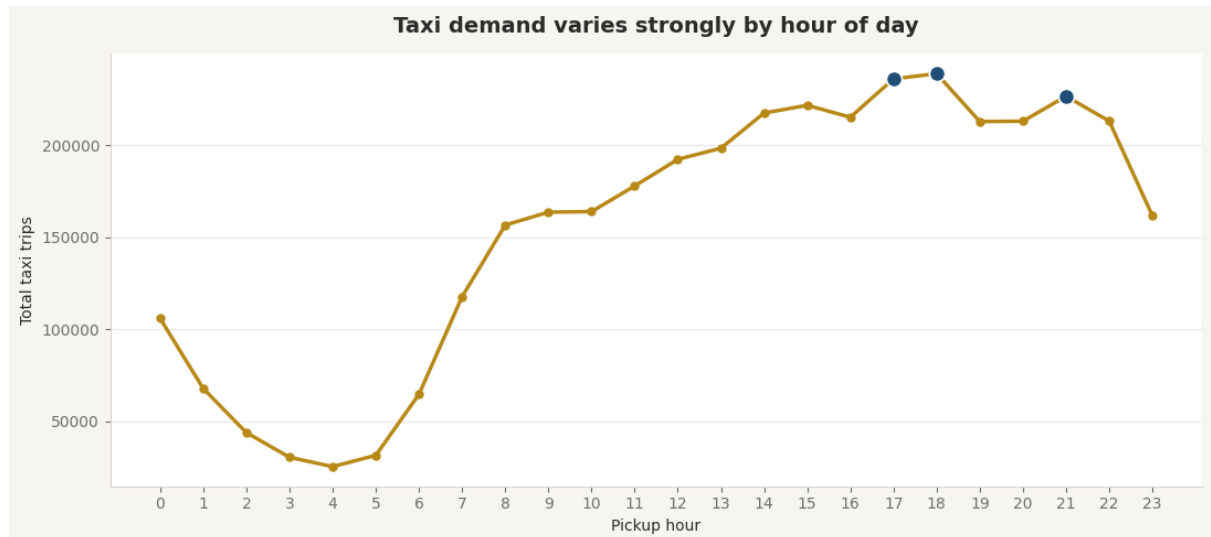
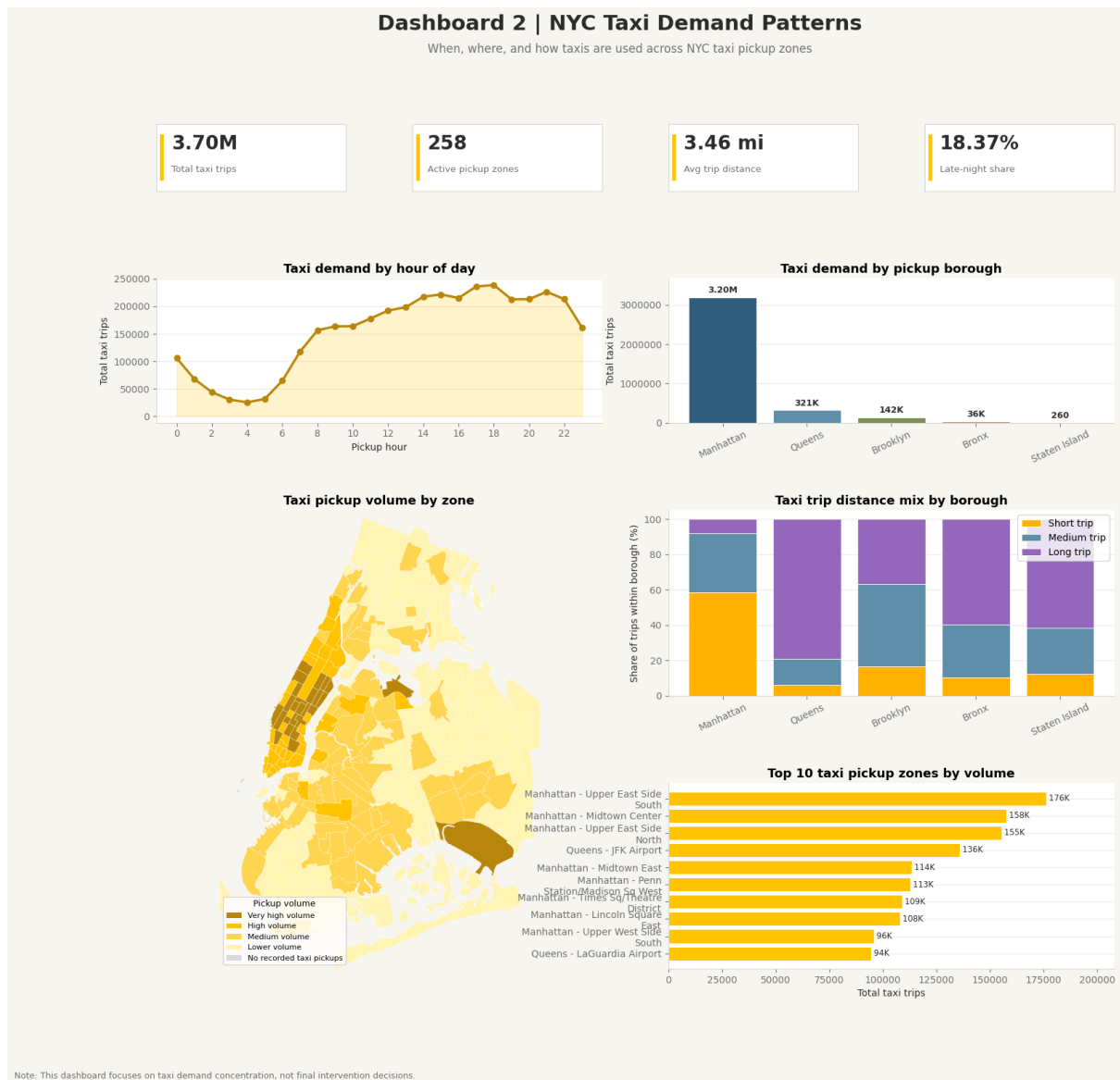


Figure 4: Taxi Demand by Hour of Day

## b) Issue identification: taxi patterns reveal different mobility needs



The issue-identification analysis shows that taxi demand should not be interpreted as one single problem. High taxi volume, late-night reliance, short-trip concentration and long-distance travel each suggest different planning responses.

Trip-distance patterns revealed different intervention needs by borough. Manhattan had the strongest short-trip pattern, with 58.58% of taxi trips classified as short trips. This supports further investigation into short-trip alternatives such as bus or Citi Bike where safe and accessible options exist.

By contrast, Queens had the strongest long-trip pattern, with 79.27% of trips classified as long trips. Bronx and Staten Island also showed stronger long-trip patterns than Manhattan. For Queens, this pattern is likely influenced by airport-related travel because JFK Airport and LaGuardia Airport were among the highest-volume pickup zones. This distinction matters because short-trip Manhattan zones are more relevant for alternative-mode review, while long trip outer-borough or airport-related zones are

more relevant for public transport connectivity review. Detailed distance-pattern outputs are provided in [Appendix H](#).

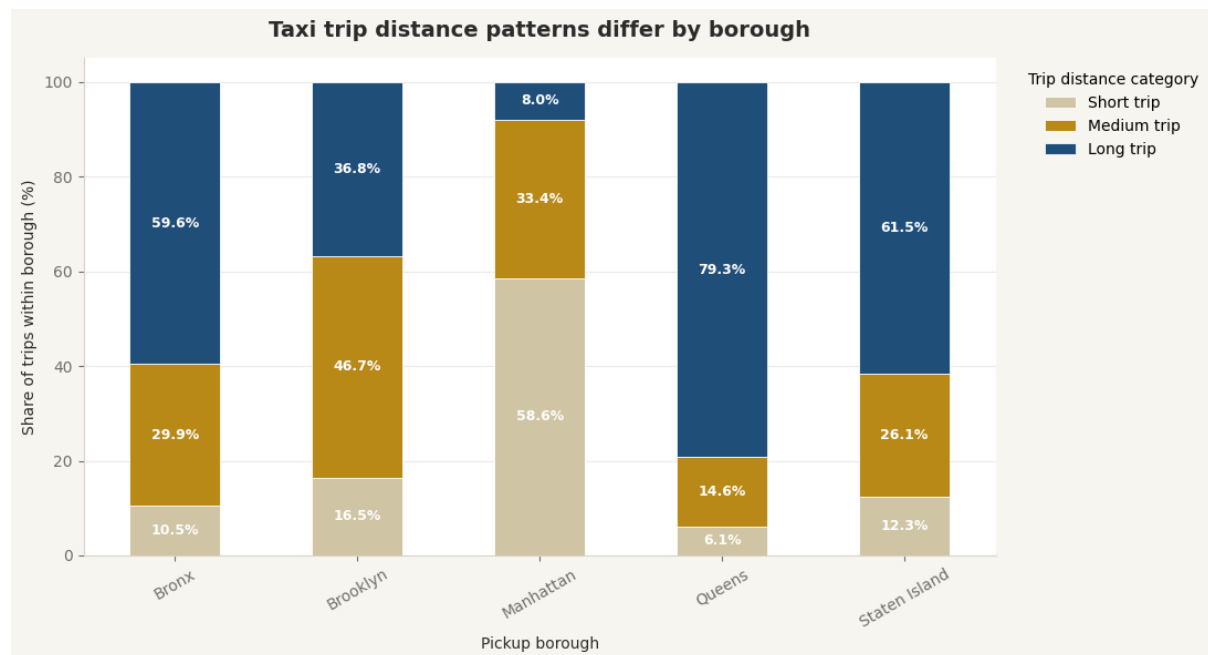


Figure 5: Taxi Trip Distance Profile by Pickup Borough

Late-night taxi demand also reveals a separate mobility signal. Although Manhattan had the largest number of late-night trips, late-night share showed that some non-Manhattan areas also relied heavily on taxis outside regular daytime and peak periods. This does not prove that subway or bus services are unavailable, but it identifies candidate areas where late-night public transport coverage, frequency, safety and accessibility may deserve further review. Detailed late-night zone outputs are provided in [Appendix E](#).

Supporting mode comparisons added wider transport context. The taxi-to-subway ratio was highest during Manhattan late night, followed by Manhattan off peak and Queens late night, suggesting taxi activity was relatively stronger compared with subway ridership outside standard peak periods. However, this does not prove that subway service was unavailable or inadequate; it only highlights areas and times for further review. The full taxi-subway comparison table is provided in [Appendix E](#).

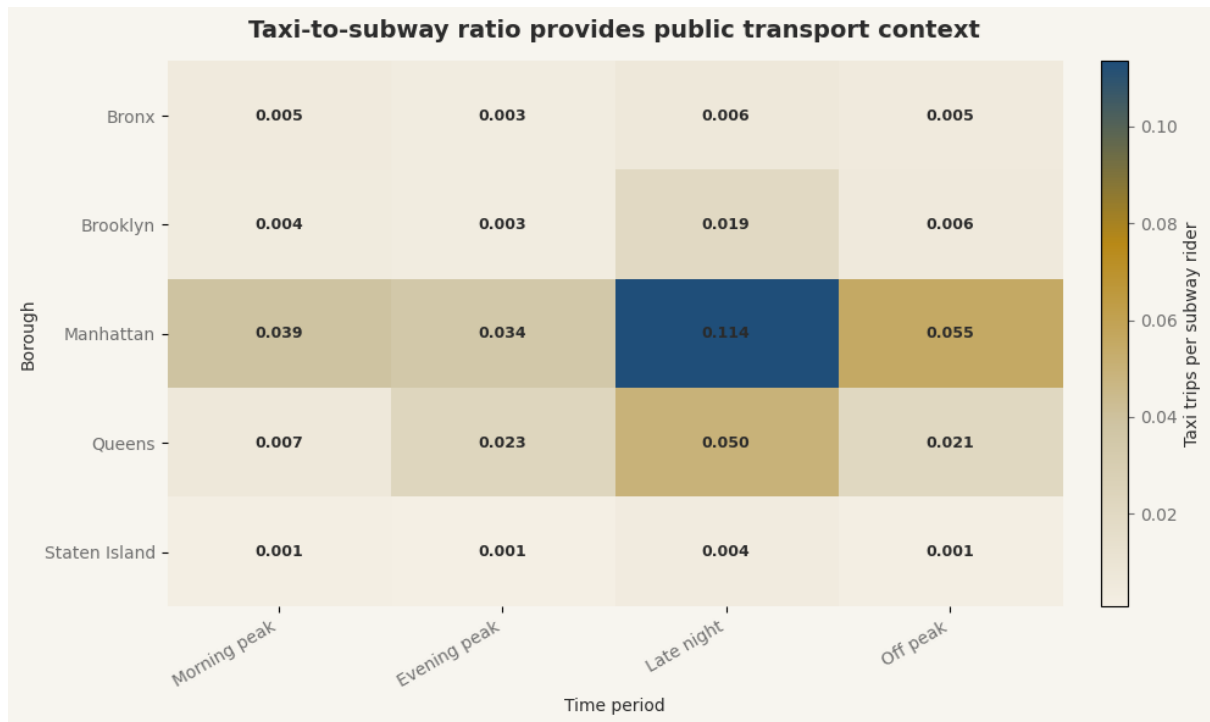


Figure 6: Taxi-to-Subway Ratio by Borough and Time Period

The taxi-to-bus comparison showed relatively strong Manhattan taxi activity in some periods, especially late night. However, this comparison is limited because bus data is route-based, and the borough variable was created from route prefixes rather than exact boarding locations. Therefore, it is used only as broad context, not precise evidence of bus service gaps. The full taxi-bus comparison and route-proxy outputs are provided in [Appendix E](#).

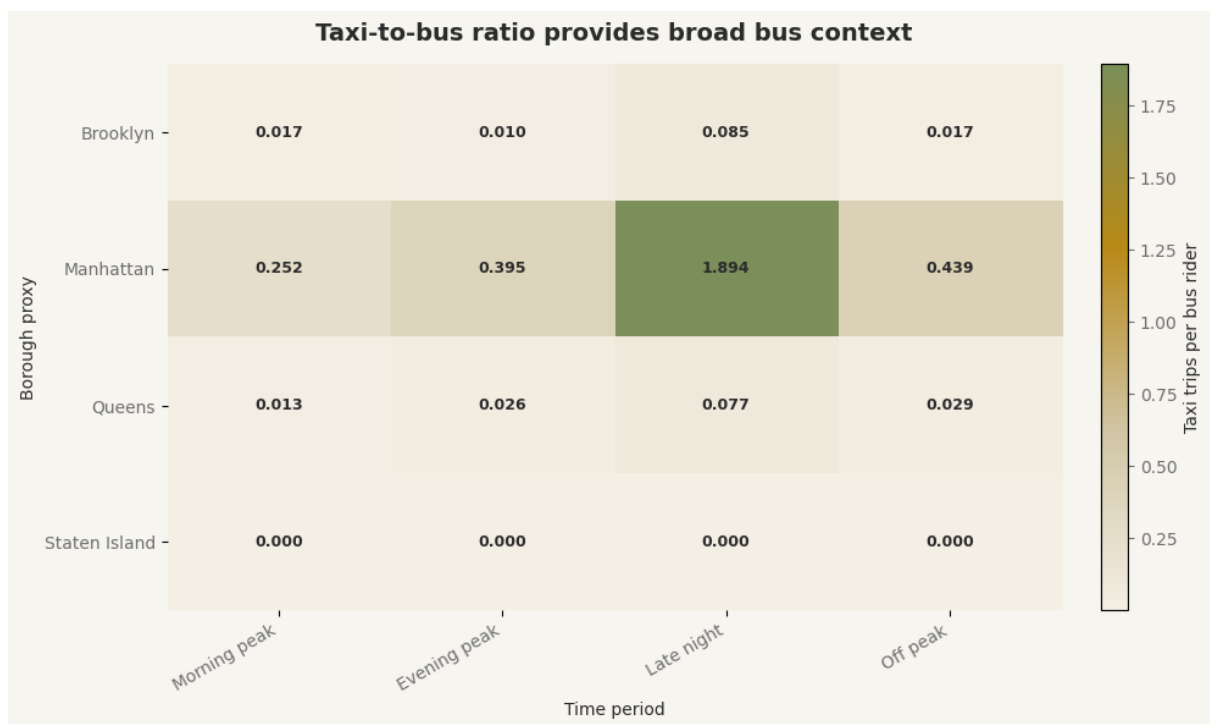
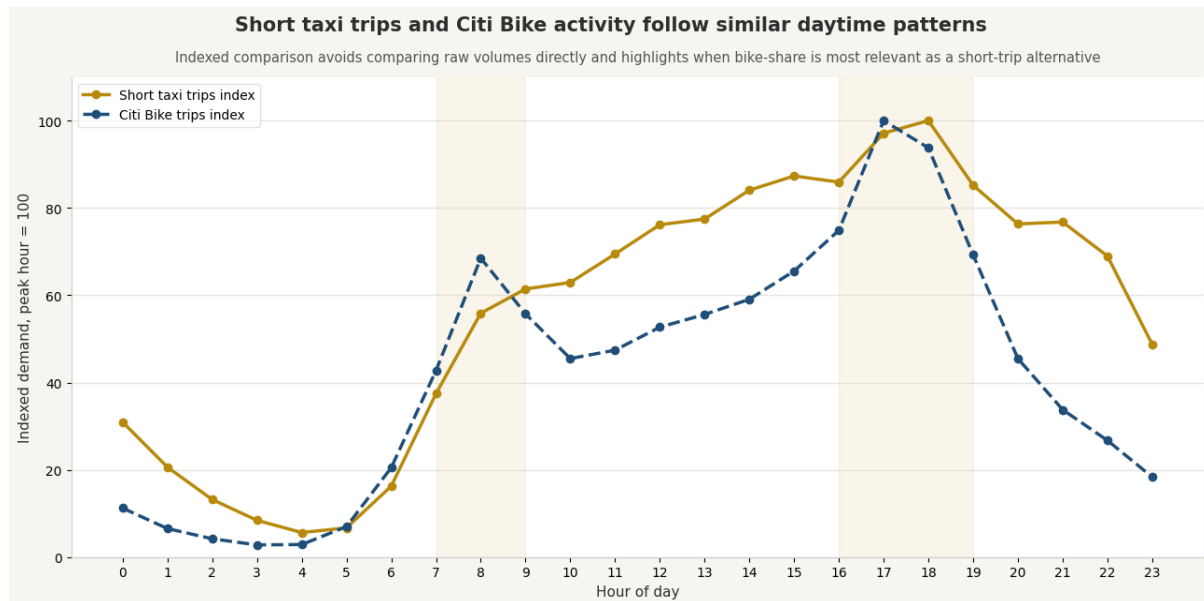


Figure 7: Taxi-to-Bus Ratio by Borough Proxy and Time Period



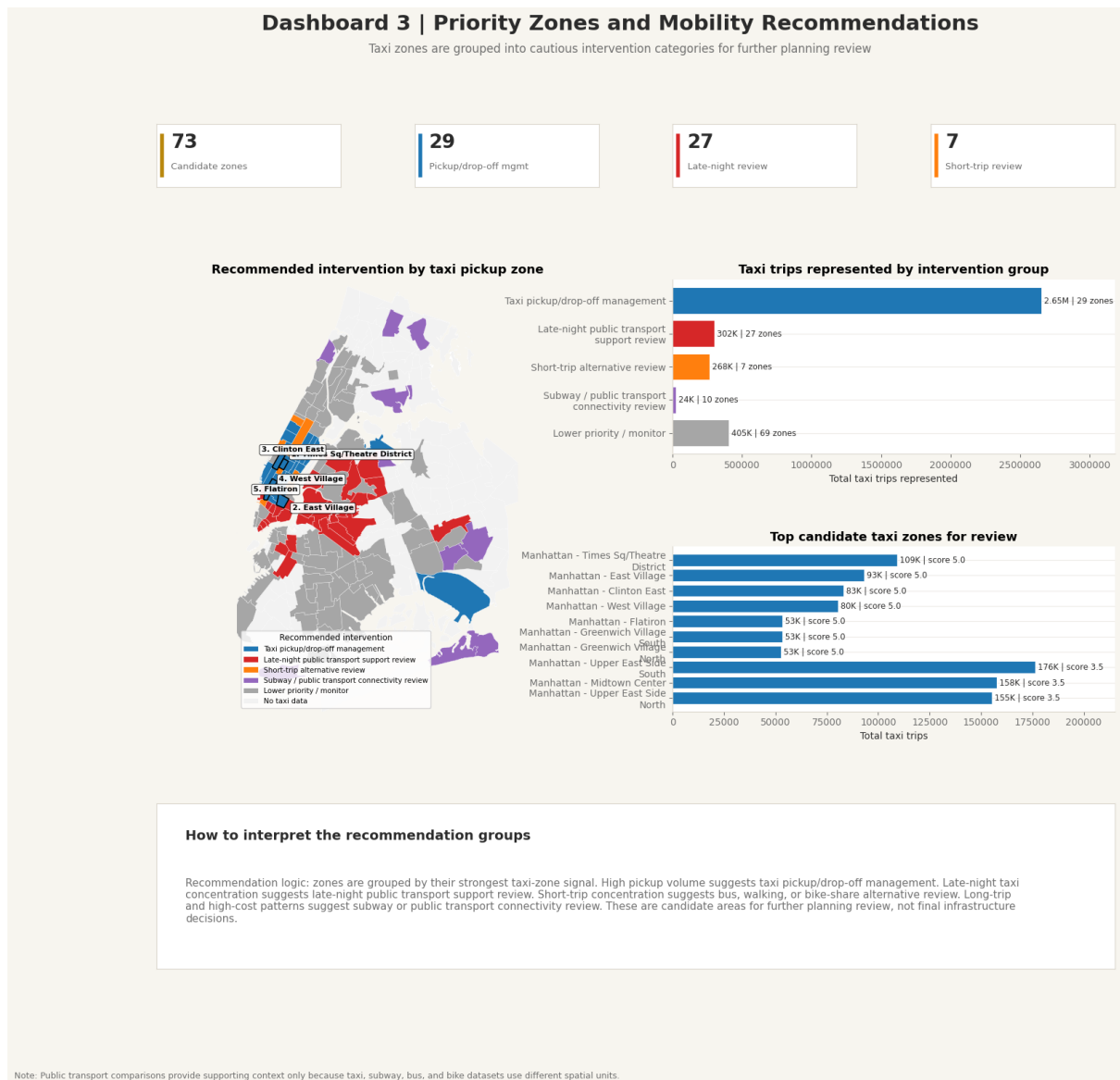
Citi Bike activity was used to support the short-trip analysis. Figure 8 compares short taxi trips and Citi Bike trips by hour using indexed values, where each mode's peak hour equals 100. This avoids directly comparing trip volumes and instead compares hourly demand patterns. Both modes showed strong evening activity, suggesting that some short-taxi demand occurs when bike-share use is also high. This may support further investigation into bike-share or other short-trip alternatives. However, it does not prove that Citi Bike can replace taxi trips, as cycling suitability depends on station coverage, safety, weather, road infrastructure and passenger needs. The full taxi-bike hourly comparison is provided [Appendix E](#).



*Figure 8: Indexed Comparison of Short Taxi Trips and Citi Bike Activity by Hour*

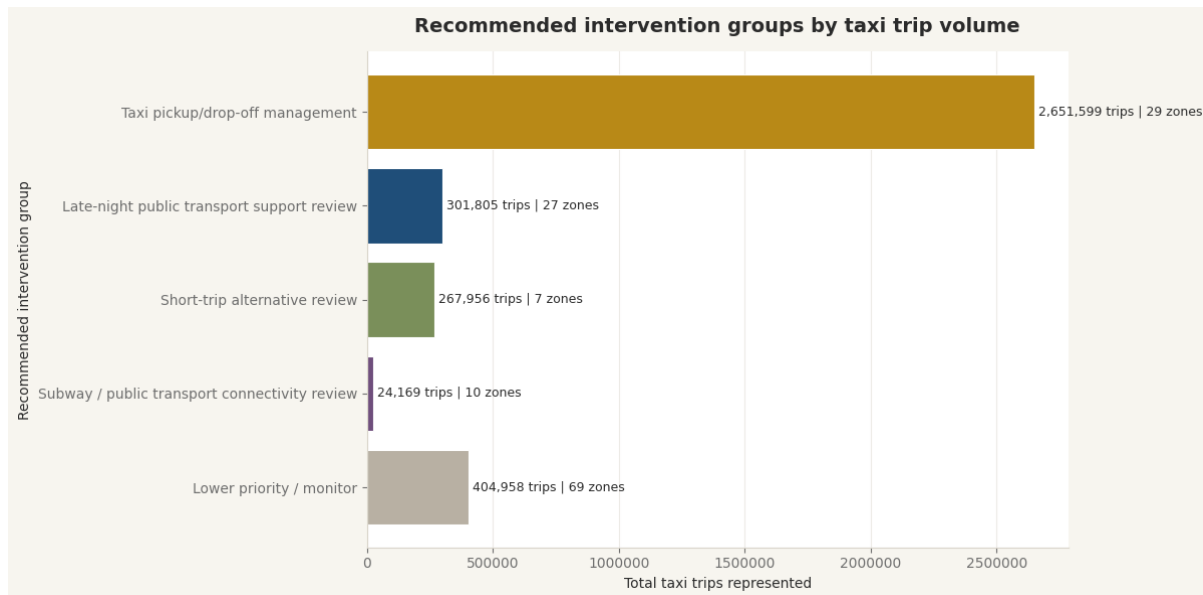
Together, the issue-identification findings show that different taxi patterns point to different mobility needs. High-volume areas suggest taxi pickup/drop-off management. Late-night taxi reliance suggests public transport support review. Short-trip concentration suggests alternative-mode investigation. Long-trip and higher-cost patterns suggest public transport connectivity review.

### c) Recommendations: intervention priorities should match the demand signal



The recommendation analysis translates the engineered taxi metrics into intervention priority groups. The key finding is that zones should not be ranked by taxi volume alone. Instead, the intervention type should match the strongest demand signal observed in each zone.

The final SQL priority table classified zones using four signals: high taxi volume, late-night concentration, short-trip concentration and connectivity. High-volume zones used a percentile threshold, while the other signals used high-share thresholds with minimum trip-count rules to avoid small-sample bias. Taxi volume received the highest weight because it affects the most trips. Late-night and short-trip signals were also weighted strongly as they indicate service-review or alternative-mode opportunities, while connectivity was treated as a screening signal requiring further evidence.



*Figure 9: Recommended Intervention Groups by Taxi Trip volume*

As shown in Figure 9, the largest intervention group was taxi pickup/drop-off management, covering 29 zones and 2,651,599 trips. These zones had high taxi volume and an average short trip share of 57.30%, suggesting strong curbside and operational pressure in dense-activity areas. Suitable interventions may include taxi stand review, pickup/drop-off zone planning and congestion management.

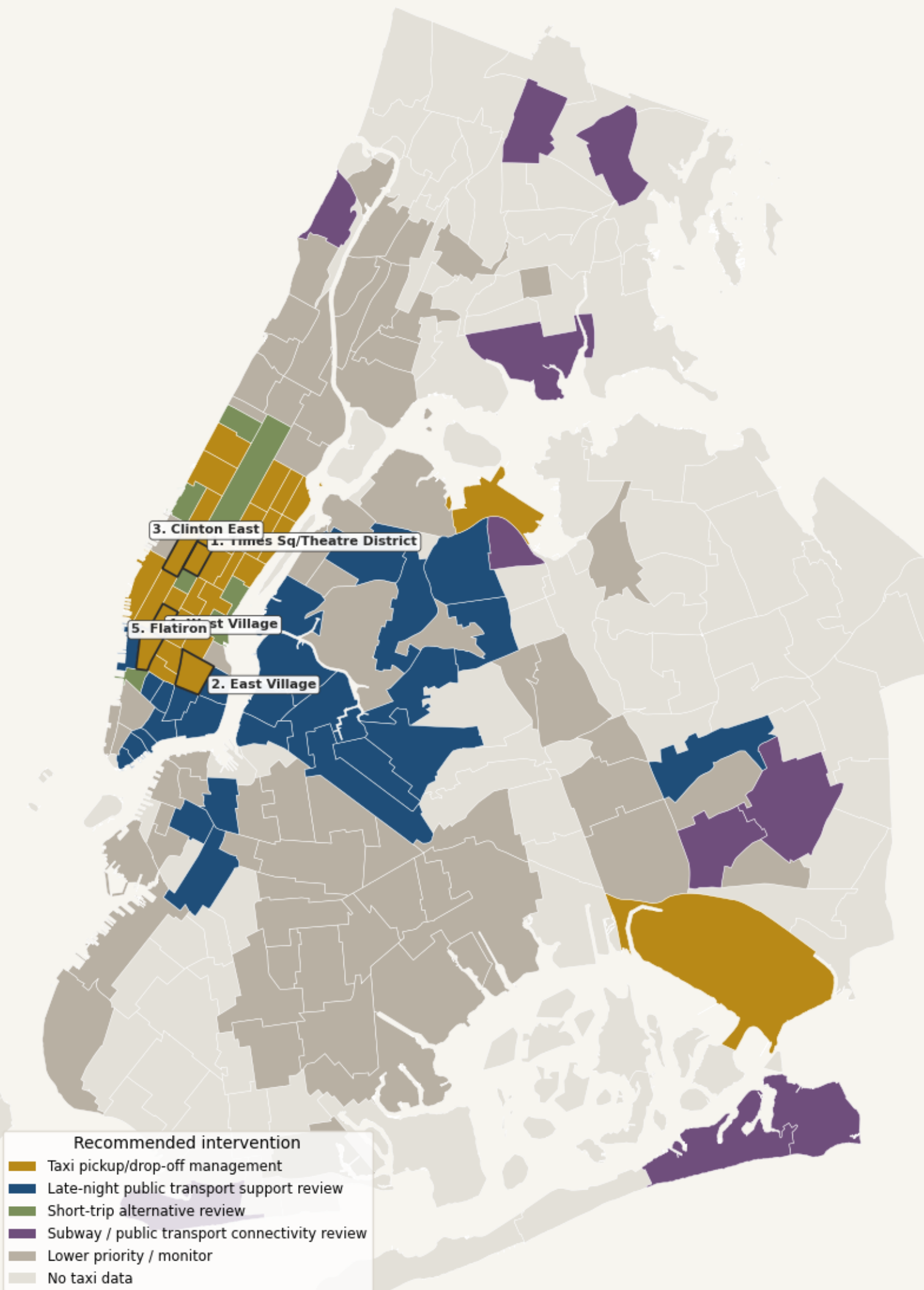
The second group was late-night public transport support review, with 27 zones and 301,805 trips. This group had the highest average late-night share at 35.96%, indicating that it was identified mainly because of late night activity rather than total volume. These zones should be reviewed for late-night mobility coverage, but taxi demand alone does not prove that public transport is unavailable.

The third group was short trip alternative review, with 7 zones and 267,956 trips. Its high short-trip share of 57.98% and low long-trip share of 8.00% suggest potential for further investigation into Citi Bike, walking or bus alternatives. However, any recommendation should consider accessibility, safety, weather and station or route coverage.

The fourth group was subway or public transport connectivity review, with 10 zones and 24,169 trips. Although trip volume was lower, this group had the highest average distance at 10.76 miles and long-trip share at 75.40%, suggesting longer or less direct travel needs. These zones should be reviewed for connectivity patterns, including airport access, cross-borough travel and limited direct public transport options.

The remaining 69 zones were classified as lower priority or monitor, covering 404,958 trips. They did not show strong enough demand signals for immediate intervention, so monitoring is appropriate, especially because the analysis is based on one month of data and should be validated with additional data and local planning knowledge.

# Recommended intervention by NYC taxi pickup zone



## VII. Conclusion

This report shows that NYC taxi demand in April 2026 was concentrated in specific places and times, especially high volume Manhattan activity zones and late afternoon/evening periods. The analysis also found different demand signals across zones, including short-trip pressure, late-night reliance and longer distance or airport-related travel.

The main recommendation is to apply targeted interventions based on these signals. High volume Manhattan zones, such as Midtown, Upper East Side, Penn Station/Madison Square West and Times Square, should be reviewed for taxi pickup/drop-off and curbside management. Late-night taxi reliant zones should be reviewed for public transport support, short trip zones for walking, bus or Citi Bike alternatives, and long-distance or airport-related zones, such as JFK, LaGuardia and parts of Queens, for connectivity improvements.

These recommendations should be treated as candidate areas for further planning review, not final infrastructure decisions. Further validation should use additional months of data, service frequency, travel times, accessibility needs and local planning context.

## References

- Citi Bike. (2025). *Citi Bike System Data | Citi Bike NYC*. Citibikenyc.com.  
<https://citibikenyc.com/system-data>
- Harvard Kennedy School. (2014, December 16). *Case Study New York City Taxis | Data-Smart City Solutions*. Harvard.edu.  
<https://datasmart.hks.harvard.edu/news/article/case-study-new-york-city-taxis-596>
- Metropolitan Transportation Authority. (2018). *MTA Subway Hourly Ridership: Beginning 2025*. Ny.gov. [https://data.ny.gov/Transportation/MTA-Subway-Hourly-Ridership-Beginning-2025/5wq4-mkjj/about\\_data](https://data.ny.gov/Transportation/MTA-Subway-Hourly-Ridership-Beginning-2025/5wq4-mkjj/about_data)
- Metropolitan Transportation Authority. (2024, November 21). *MTA Bus Hourly Ridership: Beginning 2025*. Ny.gov. [https://data.ny.gov/Transportation/MTA-Bus-Hourly-Ridership-Beginning-2025/gxb3-akrn/about\\_data](https://data.ny.gov/Transportation/MTA-Bus-Hourly-Ridership-Beginning-2025/gxb3-akrn/about_data)
- MTA. (2026). *New York City Bus Schedules*. MTA. <https://www.mta.info/schedules/bus>
- New York City Department of Transportation. (2019). *Mobility Report*.  
<https://www.nyc.gov/html/dot/downloads/pdf/mobility-report-singlepage-2019.pdf>
- TLC. (n.d.). *TLC Trip Record Data -TLC*. Wwww.nyc.gov.  
<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- TLC. (2018). *NYC Taxi Zones*. Cityofnewyork.us.  
<https://data.cityofnewyork.us/Transportation/NYC-Taxi-Zones/8meu-9t5y>
- TLC. (2025). *TLC Trip Records User Guide*.  
[https://www.nyc.gov/assets/tlc/downloads/pdf/trip\\_record\\_user\\_guide.pdf](https://www.nyc.gov/assets/tlc/downloads/pdf/trip_record_user_guide.pdf)

# Appendices

## Appendix A: Raw Data Schema and Sample Records

Appendix A.1 Taxi Zone Lookup Schema and Sample Records

|   | column       | dtype         | missing_values          | missing_pct  |
|---|--------------|---------------|-------------------------|--------------|
| 0 | LocationID   | int64         | 0                       | 0.00         |
| 1 | Borough      | object        | 1                       | 0.38         |
| 2 | Zone         | object        | 1                       | 0.38         |
| 3 | service_zone | object        | 2                       | 0.75         |
|   | LocationID   | Borough       | Zone                    | service_zone |
| 0 | 1            | EWR           | Newark Airport          | EWR          |
| 1 | 2            | Queens        | Jamaica Bay             | Boro Zone    |
| 2 | 3            | Bronx         | Allerton/Pelham Gardens | Boro Zone    |
| 3 | 4            | Manhattan     | Alphabet City           | Yellow Zone  |
| 4 | 5            | Staten Island | Arden Heights           | Boro Zone    |



Appendix A.2 Subway Schema and Sample Records

|    | column                  | dtype        | missing_values     | missing_pct                |         |                |                              |           |           |          |
|----|-------------------------|--------------|--------------------|----------------------------|---------|----------------|------------------------------|-----------|-----------|----------|
| 0  | transit_timestamp       | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
| 1  | transit_mode            | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
| 2  | station_complex_id      | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
| 3  | station_complex         | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
| 4  | borough                 | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
| 5  | payment_method          | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
| 6  | fare_class_category     | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
| 7  | ridership               | float64      | 0                  | 0.0                        |         |                |                              |           |           |          |
| 8  | transfers               | float64      | 0                  | 0.0                        |         |                |                              |           |           |          |
| 9  | latitude                | float64      | 0                  | 0.0                        |         |                |                              |           |           |          |
| 10 | longitude               | float64      | 0                  | 0.0                        |         |                |                              |           |           |          |
| 11 | georeference            | object       | 0                  | 0.0                        |         |                |                              |           |           |          |
|    | transit_timestamp       | transit_mode | station_complex_id | station_complex            | borough | payment_method | fare_class_category          | ridership | transfers | latitude |
| 0  | 2026-04-01T00:00:00.000 | subway       | 1                  | Astoria-Ditmars Blvd (N,W) | Queens  | metrocard      | Metrocard - Unlimited 30-Day | 1.0       | 0.0       | 40.775   |
| 1  | 2026-04-01T00:00:00.000 | subway       | 1                  | Astoria-Ditmars Blvd (N,W) | Queens  | omny           | OMNY - Other                 | 8.0       | 0.0       | 40.775   |
| 2  | 2026-04-01T00:00:00.000 | subway       | 1                  | Astoria-Ditmars Blvd (N,W) | Queens  | omny           | OMNY - Fair Fare             | 2.0       | 0.0       | 40.775   |
| 3  | 2026-04-01T00:00:00.000 | subway       | 1                  | Astoria-Ditmars Blvd (N,W) | Queens  | metrocard      | Metrocard - Other            | 2.0       | 0.0       | 40.775   |
| 4  | 2026-04-01T00:00:00.000 | subway       | 1                  | Astoria-Ditmars Blvd (N,W) | Queens  | omny           | OMNY - Full Fare             | 33.0      | 0.0       | 40.775   |

### Appendix A.3 Bus Schema and Sample Records

|   | transit_timestamp       | bus_route | payment_method | fare_class_category         | ridership | transfers |
|---|-------------------------|-----------|----------------|-----------------------------|-----------|-----------|
| 0 | 2026-04-01T00:00:00.000 | B1        | omny           | Metrocard - Fair Fare       | 0         | 0         |
| 1 | 2026-04-01T00:00:00.000 | B1        | omny           | Metrocard - Students        | 0         | 0         |
| 2 | 2026-04-01T00:00:00.000 | B1        | omny           | OMNY - Students             | 1         | 0         |
| 3 | 2026-04-01T00:00:00.000 | B1        | omny           | Metrocard - Other           | 0         | 0         |
| 4 | 2026-04-01T00:00:00.000 | B1        | omny           | OMNY - Seniors & Disability | 3         | 1         |

|   | column              | dtype  | missing_values | missing_pct |
|---|---------------------|--------|----------------|-------------|
| 0 | transit_timestamp   | object | 0              | 0.0         |
| 1 | bus_route           | object | 0              | 0.0         |
| 2 | payment_method      | object | 0              | 0.0         |
| 3 | fare_class_category | object | 0              | 0.0         |
| 4 | ridership           | int64  | 0              | 0.0         |
| 5 | transfers           | int64  | 0              | 0.0         |

### Appendix A.4 Citi Bike Schema and Sample Records

|    | column_name        | column_type | null | key  | default | extra |
|----|--------------------|-------------|------|------|---------|-------|
| 0  | ride_id            | VARCHAR     | YES  | None | None    | None  |
| 1  | rideable_type      | VARCHAR     | YES  | None | None    | None  |
| 2  | started_at         | VARCHAR     | YES  | None | None    | None  |
| 3  | ended_at           | VARCHAR     | YES  | None | None    | None  |
| 4  | start_station_name | VARCHAR     | YES  | None | None    | None  |
| 5  | start_station_id   | VARCHAR     | YES  | None | None    | None  |
| 6  | end_station_name   | VARCHAR     | YES  | None | None    | None  |
| 7  | end_station_id     | VARCHAR     | YES  | None | None    | None  |
| 8  | start_lat          | VARCHAR     | YES  | None | None    | None  |
| 9  | start_lng          | VARCHAR     | YES  | None | None    | None  |
| 10 | end_lat            | VARCHAR     | YES  | None | None    | None  |
| 11 | end_lng            | VARCHAR     | YES  | None | None    | None  |
| 12 | member_casual      | VARCHAR     | YES  | None | None    | None  |

CSV files inside bike zip: ['202604-citibike-tripdata-part1.csv', '202604-citibike-tripdata-part2.csv', '202604-citibike-tripdata-part3.csv', '202604-citibike-tripdata-part4.csv']  
Reading: 202604-citibike-tripdata-part1.csv  
Reading: 202604-citibike-tripdata-part2.csv  
Reading: 202604-citibike-tripdata-part3.csv  
Reading: 202604-citibike-tripdata-part4.csv  
Extracted and combined bike CSV to: /content/drive/MyDrive/nyc\_mobility\_pipeline/raw\_data/bike/citibike\_tripdata\_2026-04.csv  
Bike raw shape: (3860371, 13)

|   | ride_id          | rideable_type | started_at              | ended_at                | start_station_name        | start_station_id | end_station_name        | end_station_id | start_lat | start_lng  | end_lat   | end_lng    |
|---|------------------|---------------|-------------------------|-------------------------|---------------------------|------------------|-------------------------|----------------|-----------|------------|-----------|------------|
| 0 | C88217D5C42993C6 | classic_bike  | 2026-04-06 18:55:38.804 | 2026-04-06 19:04:03.989 | E 88 St & Park Ave        | 7293.10          | E 106 St & 1 Ave        | 7456.03        | 40.781411 | -73.965411 | 40.781411 | -73.965411 |
| 1 | 44E5BC08D0125145 | classic_bike  | 2026-04-09 01:48:51.203 | 2026-04-09 01:58:40.980 | Broadway & W 61 St        | 7014.12          | West End Ave & W 60 St  | 7059.08        | 40.770030 | -73.981530 | 40.770030 | -73.981530 |
| 2 | D4EA18EA685E8D59 | electric_bike | 2026-04-04 03:21:33.088 | 2026-04-04 03:33:35.930 | Delancey St & Eldridge St | 5414.07          | Lexington Ave & E 24 St | 6089.07        | 40.719383 | -73.981530 | 40.719383 | -73.981530 |
| 3 | 5BA741D2CAC6AD6A | electric_bike | 2026-04-11 11:58:36.715 | 2026-04-11 14:07:16.782 | Cherry St                 | 5181.04          | E 13 St & 2 Ave         | 5820.08        | 40.712199 | -73.981530 | 40.712199 | -73.981530 |
| 4 | 0804FB9898A542A3 | classic_bike  | 2026-04-07 09:37:46.372 | 2026-04-07 09:45:39.175 | Delancey St & Eldridge St | 5414.07          | E 13 St & 2 Ave         | 5820.08        | 40.719383 | -73.981530 | 40.719383 | -73.981530 |

## Appendix B: Issue Severity Classification Rules

| Severity | Rule used in pipeline                           | Interpretation                                                                         |
|----------|-------------------------------------------------|----------------------------------------------------------------------------------------|
| None     | 0 affected rows                                 | No cleaning required for that issue.                                                   |
| Low      | Supporting datasets: <1%; Taxi: <2%             | Minor issue; document, flag, or handle only in relevant analysis.                      |
| Medium   | 1-<10% for supporting datasets; 2-<10% for taxi | Meaningful issue; apply cleaning or selective exclusion if it affects analysis fields. |
| High     | ≥10%                                            | Major issue; must be addressed or clearly justified before analysis.                   |

## Appendix C: Supporting Dataset Audit Results

### Appendix C.1 Taxi Zone Lookup Audit Summary

|   | Borough       | zone_count |
|---|---------------|------------|
| 0 | Queens        | 69         |
| 1 | Manhattan     | 69         |
| 2 | Brooklyn      | 61         |
| 3 | Bronx         | 43         |
| 4 | Staten Island | 20         |
| 5 | N/A           | 1          |
| 6 | EWR           | 1          |
| 7 | Unknown       | 1          |

|   | dataset          | issue                                         | affected_rows | affected_pct | severity | recommended_action                                                                                  |
|---|------------------|-----------------------------------------------|---------------|--------------|----------|-----------------------------------------------------------------------------------------------------|
| 0 | taxi_zone_lookup | Duplicate or missing key zone fields          | 0             | 0.00%        | None     | No cleaning required if affected rows are zero; otherwise review before joining with taxi trip data |
| 1 | taxi_zone_lookup | Special borough categories: N/A, EWR, Unknown | 3             | 1.13%        | Medium   | Keep if valid in the TLC lookup, but treat carefully in borough-level analysis                      |

### Appendix C.2 Subway Audit Summary

|   | borough       | row_count | total_ridership |
|---|---------------|-----------|-----------------|
| 0 | Manhattan     | 531237    | 64145902.0      |
| 1 | Brooklyn      | 606558    | 25406445.0      |
| 2 | Queens        | 304554    | 16544899.0      |
| 3 | Bronx         | 275749    | 7753567.0       |
| 4 | Staten Island | 6754      | 193152.0        |

| index | dataset | issue                                 | affected_rows | affected_pct | severity | recommended_action                                                |
|-------|---------|---------------------------------------|---------------|--------------|----------|-------------------------------------------------------------------|
| 0     | subway  | Missing values in key fields          | 0             | 0.00%        | None     | No missing-value cleaning required for key subway fields          |
| 1     | subway  | Negative ridership or transfer values | 0             | 0.00%        | None     | No negative-value cleaning required                               |
| 2     | subway  | Date coverage                         | 0             | 0.00%        | None     | Dataset covers April 2026 and can be used for time-based analysis |
| 3     | subway  | Borough coverage                      | N/A           | N/A          | None     | Dataset covers 5 boroughs and 428 station complexes               |

### Appendix C.3 Bus Audit Summary

|    | route_prefix | row_count | unique_routes | total_ridership |
|----|--------------|-----------|---------------|-----------------|
| 0  | Q            | 1866240   | 108           | 11281060.0      |
| 1  | B            | 1226880   | 71            | 7528222.0       |
| 2  | M            | 794880    | 46            | 7099307.0       |
| 3  | BX           | 881280    | 51            | 4345798.0       |
| 4  | S            | 535680    | 31            | 1112653.0       |
| 5  | SIM          | 535680    | 31            | 692286.0        |
| 6  | BXM          | 190080    | 11            | 349451.0        |
| 7  | QM           | 535680    | 31            | 286232.0        |
| 8  | X            | 535680    | 31            | 124477.0        |
| 9  | BM           | 86400     | 5             | 90678.0         |
| 10 | J            | 34560     | 2             | 479.0           |
| 11 | L            | 51840     | 3             | 46.0            |
| 12 | T            | 51840     | 3             | 5.0             |
| 13 | F            | 17280     | 1             | 0.0             |
| 14 | D            | 34560     | 2             | 0.0             |

|   | dataset | issue                                     | affected_rows | affected_pct | severity | recommended_action                                                        |
|---|---------|-------------------------------------------|---------------|--------------|----------|---------------------------------------------------------------------------|
| 0 | bus     | Missing values in key fields              | 0             | 0.00%        | None     | No missing-value cleaning required for key bus fields                     |
| 1 | bus     | Negative ridership or transfer values     | 0             | 0.00%        | None     | No negative-value cleaning required                                       |
| 2 | bus     | Very low or zero ridership route prefixes | 155,520       | 2.11%        | Medium   | Review very low-ridership prefixes before using them in demand comparison |

### Appendix C.4 Citi Bike Audit Summary

| Metric                    | Value               |
|---------------------------|---------------------|
| Index                     | 0                   |
| Total Rows                | 3,860,371           |
| Unique Ride IDs           | 3,860,371           |
| Duplicate Ride IDs        | 0                   |
| Min Started At            | 2026-04-01 0:00:03  |
| Max Started At            | 2026-04-30 23:59:59 |
| Missing Started At        | 0                   |
| Missing Ended At          | 0                   |
| Invalid Time Order        | 0                   |
| Missing Start Station     | 1,939               |
| Missing End Station       | 9,886               |
| Missing Start Coordinates | 1,939               |
| Missing End Coordinates   | 10,473              |
| Unique Bike Types         | 2                   |
| Unique User Types         | 2                   |

| index | dataset | issue                              | affected_rows | affected_pct | severity | recommended_action                                                                                  |
|-------|---------|------------------------------------|---------------|--------------|----------|-----------------------------------------------------------------------------------------------------|
| 0     | bike    | Duplicate ride IDs                 | 0             | 0.00%        | None     | No duplicate cleaning required                                                                      |
| 1     | bike    | Missing or invalid trip timestamps | 0             | 0.00%        | None     | No timestamp cleaning required for trip duration analysis                                           |
| 2     | bike    | Missing station information        | 11,825        | 0.31%        | Low      | Keep for overall bike demand analysis, but exclude missing station rows from station-level analysis |
| 3     | bike    | Missing coordinate information     | 12,412        | 0.32%        | Low      | Keep for general demand analysis, but exclude missing coordinate rows from map-based analysis       |
| 4     | bike    | Date coverage                      | 0             | 0.00%        | None     | Dataset covers April 2026 and can be used for time-based analysis                                   |
| 5     | bike    | Category coverage                  | N/A           | N/A          | None     | Dataset covers 2 bike types and 2 user types                                                        |

## Appendix D: Cleaning and Transformation Results.

### Appendix D.1 Taxi Feature Engineering Results

| <b>Metric</b>      | <b>Count</b> | <b>% of Total Trips</b> |
|--------------------|--------------|-------------------------|
| Morning Peak Trips | 437,826      | 11.85%                  |
| Evening Peak Trips | 903,164      | 24.44%                  |
| Late Night Trips   | 679,038      | 18.37%                  |
| Short Trips        | 1,919,436    | 51.93%                  |
| Medium Trips       | 1,191,314    | 32.23%                  |
| Long Trips         | 585,251      | 15.84%                  |

#### Appendix D.2 Subway Cleaning Results

| <b>Metric</b>            | <b>Value</b>        |
|--------------------------|---------------------|
| Cleaned Rows             | 1,724,851           |
| Min Transit Timestamp    | 2026-04-01 0:00:00  |
| Max Transit Timestamp    | 2026-04-30 23:00:00 |
| Unique Station Complexes | 428                 |
| Unique Boroughs          | 5                   |
| Total Ridership          | 114,043,964         |
| Total Transfers          | 6,181,908           |

#### Appendix D.3 Bus Cleaning and Route-Proxy Results

| <b>Metric</b>                | <b>Value</b>        |
|------------------------------|---------------------|
| Cleaned Rows                 | 7,378,560           |
| Min Transit Timestamp        | 2026-04-01 0:00:00  |
| Max Transit Timestamp        | 2026-04-30 23:00:00 |
| Unique Bus Routes            | 427                 |
| Unique Route Prefixes        | 15                  |
| Unique Route Borough Proxies | 6                   |
| Total Ridership              | 32,910,694          |
| Total Transfers              | 8,840,500           |

|   | route_borough_proxy  | rows    | unique_routes | total_ridership |
|---|----------------------|---------|---------------|-----------------|
| 0 | Queens               | 1866240 | 108           | 11281060.0      |
| 1 | Brooklyn             | 1226880 | 71            | 7528222.0       |
| 2 | Manhattan            | 794880  | 46            | 7099307.0       |
| 3 | Other/Unknown        | 1797120 | 104           | 4820256.0       |
| 4 | Staten Island        | 1071360 | 62            | 1804939.0       |
| 5 | Express/Interborough | 622080  | 36            | 376910.0        |

#### Appendix D.5 Citi Bike Cleaning Results

| Metric                            | Value               |
|-----------------------------------|---------------------|
| Cleaned Rows                      | 3,859,605           |
| Unique Ride IDs                   | 3,859,605           |
| Min Started At                    | 2026-04-01 0:00:03  |
| Max Started At                    | 2026-04-30 23:59:59 |
| Average Ride Duration (Minutes)   | 12.74               |
| Rows with Missing Station Info    | 10,856              |
| Rows with Missing Coordinate Info | 11,466              |
| Unique Bike Types                 | 2                   |
| Unique User Types                 | 2                   |

## Appendix E: SQL Summary Tables

### Appendix E.1 taxi\_borough\_summary

| pickup_borough | total_taxi_trips | yellow_taxi_trips | green_taxi_trips | avg_trip_distance_miles | avg_total_amount_usd | avg_trip_duration_minutes | late_night_trip_share_pct | short_trip_share_pct |
|----------------|------------------|-------------------|------------------|-------------------------|----------------------|---------------------------|---------------------------|----------------------|
| Manhattan      | 3196309          | 3170897           | 25412            | 2.54                    | 25.74                | 15.69                     | 17.97%                    | 58.58%               |
| Queens         | 321397           | 312601            | 8796             | 11.00                   | 65.45                | 35.77                     | 20.39%                    | 6.10%                |
| Brooklyn       | 141851           | 135172            | 6679             | 5.97                    | 35.16                | 31.82                     | 24.20%                    | 16.48%               |
| Bronx          | 36085            | 34890             | 1195             | 8.52                    | 37.74                | 40.90                     | 13.39%                    | 10.49%               |
| Staten Island  | 260              | 252               | 8                | 9.73                    | 50.11                | 31.65                     | 19.62%                    | 12.31%               |



Appendix E.2 taxi\_hourly\_summary

|    | pickup_hour | total_taxi_trips | avg_trip_distance_miles | avg_total_amount_usd | short_trips | long_trips |
|----|-------------|------------------|-------------------------|----------------------|-------------|------------|
| 0  | 0           | 105792           | 3.90                    | 30.53                | 43871       | 20189      |
| 1  | 1           | 67700            | 3.53                    | 27.42                | 29215       | 10587      |
| 2  | 2           | 43702            | 3.40                    | 26.18                | 18718       | 6584       |
| 3  | 3           | 30329            | 3.81                    | 27.80                | 11952       | 5750       |
| 4  | 4           | 25240            | 5.08                    | 32.50                | 7972        | 7579       |
| 5  | 5           | 31393            | 6.05                    | 35.65                | 9489        | 11907      |
| 6  | 6           | 64557            | 5.11                    | 31.41                | 22915       | 19437      |
| 7  | 7           | 117628           | 3.94                    | 29.07                | 52937       | 23389      |
| 8  | 8           | 156591           | 3.43                    | 28.58                | 79077       | 24735      |
| 9  | 9           | 163600           | 3.31                    | 27.86                | 87049       | 24821      |
| 10 | 10          | 163896           | 3.33                    | 28.18                | 89239       | 25720      |
| 11 | 11          | 177841           | 3.35                    | 28.78                | 98510       | 27433      |
| 12 | 12          | 192323           | 3.30                    | 28.86                | 108118      | 29120      |
| 13 | 13          | 198397           | 3.43                    | 29.69                | 109924      | 31453      |
| 14 | 14          | 217494           | 3.53                    | 30.60                | 119209      | 36109      |
| 15 | 15          | 221703           | 3.48                    | 30.60                | 123999      | 35610      |
| 16 | 16          | 215235           | 3.48                    | 32.02                | 121992      | 34118      |
| 17 | 17          | 236087           | 3.10                    | 30.32                | 137909      | 30403      |
| 18 | 18          | 238932           | 2.95                    | 29.19                | 142061      | 27757      |
| 19 | 19          | 212875           | 3.16                    | 29.56                | 120989      | 27650      |
| 20 | 20          | 213067           | 3.33                    | 28.67                | 108329      | 30065      |
| 21 | 21          | 226643           | 3.40                    | 29.49                | 108861      | 33294      |
| 22 | 22          | 213128           | 3.49                    | 30.19                | 97777       | 32479      |
| 23 | 23          | 161749           | 3.78                    | 30.94                | 69229       | 29059      |

Appendix E.3 taxi\_zone\_summary

|    | pickup_location_id | pickup_borough | pickup_zone                  | pickup_service_zone | total_taxi_trips | yellow_taxi_trips | green_taxi_trips | avg_trip_distance_miles | avg_total_amount_usd | avg_trip_duration |
|----|--------------------|----------------|------------------------------|---------------------|------------------|-------------------|------------------|-------------------------|----------------------|-------------------|
| 0  | 237                | Manhattan      | Upper East Side South        | Yellow Zone         | 176,330          | 176,310           | 20               | 1.81                    | 21.90                |                   |
| 1  | 161                | Manhattan      | Midtown Center               | Yellow Zone         | 157,679          | 157,663           | 16               | 2.31                    | 27.25                |                   |
| 2  | 236                | Manhattan      | Upper East Side North        | Yellow Zone         | 155,473          | 155,176           | 297              | 2.03                    | 22.50                |                   |
| 3  | 132                | Queens         | JFK Airport                  | Airports            | 135,796          | 135,788           | 8                | 14.79                   | 77.78                |                   |
| 4  | 162                | Manhattan      | Midtown East                 | Yellow Zone         | 113,648          | 113,632           | 16               | 2.28                    | 25.98                |                   |
| 5  | 186                | Manhattan      | Penn Station/Madison Sq West | Yellow Zone         | 113,022          | 113,009           | 13               | 2.30                    | 26.42                |                   |
| 6  | 230                | Manhattan      | Times Sq/Theatre District    | Yellow Zone         | 109,175          | 109,166           | 9                | 2.89                    | 29.95                |                   |
| 7  | 142                | Manhattan      | Lincoln Square East          | Yellow Zone         | 108,049          | 108,027           | 22               | 2.23                    | 23.71                |                   |
| 8  | 239                | Manhattan      | Upper West Side South        | Yellow Zone         | 95,862           | 95,844            | 18               | 2.28                    | 23.58                |                   |
| 9  | 138                | Queens         | LaGuardia Airport            | Airports            | 94,448           | 94,441            | 7                | 9.41                    | 69.46                |                   |
| 10 | 170                | Manhattan      | Murray Hill                  | Yellow Zone         | 94,236           | 94,212            | 24               | 2.38                    | 25.68                |                   |
| 11 | 234                | Manhattan      | Union Sq                     | Yellow Zone         | 93,684           | 93,670            | 14               | 2.14                    | 25.04                |                   |
| 12 | 79                 | Manhattan      | East Village                 | Yellow Zone         | 93,298           | 93,293            | 5                | 2.54                    | 25.46                |                   |
| 13 | 163                | Manhattan      | Midtown North                | Yellow Zone         | 93,132           | 93,120            | 12               | 2.36                    | 26.55                |                   |
| 14 | 68                 | Manhattan      | East Chelsea                 | Yellow Zone         | 88,310           | 88,298            | 12               | 2.59                    | 27.68                |                   |
| 15 | 48                 | Manhattan      | Clinton East                 | Yellow Zone         | 83,191           | 83,175            | 16               | 2.59                    | 25.89                |                   |
| 16 | 141                | Manhattan      | Lenox Hill West              | Yellow Zone         | 80,686           | 80,663            | 23               | 2.11                    | 22.42                |                   |
| 17 | 249                | Manhattan      | West Village                 | Yellow Zone         | 80,404           | 80,399            | 5                | 2.29                    | 25.54                |                   |

## Appendix E.4 Short-Trip Zone Summary

|    | pickup_location_id | pickup_borough | pickup_zone                  | total_taxi_trips | short_trips | short_trip_share_pct | avg_total_amount_usd | avg_trip_distance_miles |
|----|--------------------|----------------|------------------------------|------------------|-------------|----------------------|----------------------|-------------------------|
| 0  | 237                | Manhattan      | Upper East Side South        | 176330           | 132811      | 75.32                | 21.90                | 1.81                    |
| 1  | 236                | Manhattan      | Upper East Side North        | 155473           | 107133      | 68.91                | 22.50                | 2.03                    |
| 2  | 161                | Manhattan      | Midtown Center               | 157679           | 101027      | 64.07                | 27.25                | 2.31                    |
| 3  | 162                | Manhattan      | Midtown East                 | 113648           | 73573       | 64.74                | 25.98                | 2.28                    |
| 4  | 186                | Manhattan      | Penn Station/Madison Sq West | 113022           | 70006       | 61.94                | 26.42                | 2.30                    |
| 5  | 142                | Manhattan      | Lincoln Square East          | 108049           | 68369       | 63.28                | 23.71                | 2.23                    |
| 6  | 230                | Manhattan      | Times Sq/Theatre District    | 109175           | 63796       | 58.43                | 29.95                | 2.89                    |
| 7  | 234                | Manhattan      | Union Sq                     | 93684            | 61586       | 65.74                | 25.04                | 2.14                    |
| 8  | 163                | Manhattan      | Midtown North                | 93132            | 60922       | 65.41                | 26.55                | 2.36                    |
| 9  | 239                | Manhattan      | Upper West Side South        | 95862            | 59581       | 62.15                | 23.58                | 2.28                    |
| 10 | 170                | Manhattan      | Murray Hill                  | 94236            | 58380       | 61.95                | 25.68                | 2.38                    |
| 11 | 141                | Manhattan      | Lenox Hill West              | 80686            | 53992       | 66.92                | 22.42                | 2.11                    |
| 12 | 79                 | Manhattan      | East Village                 | 93298            | 49868       | 53.45                | 25.46                | 2.54                    |
| 13 | 68                 | Manhattan      | East Chelsea                 | 88310            | 48622       | 55.06                | 27.68                | 2.59                    |
| 14 | 249                | Manhattan      | West Village                 | 80404            | 48571       | 60.41                | 25.54                | 2.29                    |
| 15 | 48                 | Manhattan      | Clinton East                 | 83191            | 48412       | 58.19                | 25.89                | 2.59                    |
| 16 | 164                | Manhattan      | Midtown South                | 70043            | 43982       | 62.79                | 26.36                | 2.41                    |
| 17 | 107                | Manhattan      | Gramercy                     | 67305            | 42179       | 62.67                | 24.89                | 2.37                    |
| 18 | 238                | Manhattan      | Upper West Side North        | 70933            | 40735       | 57.43                | 23.35                | 2.47                    |
| 19 | 140                | Manhattan      | Lenox Hill East              | 73074            | 39679       | 54.30                | 24.97                | 2.69                    |

## Appendix E.5 Long-Trip Zone Summary

|    | pickup_location_id | pickup_borough | pickup_zone                  | total_taxi_trips | long_trips | long_trip_share_pct | avg_total_amount_usd | avg_trip_distance_miles |
|----|--------------------|----------------|------------------------------|------------------|------------|---------------------|----------------------|-------------------------|
| 0  | 132                | Queens         | JFK Airport                  | 135796           | 121693     | 89.61               | 77.78                | 14.79                   |
| 1  | 138                | Queens         | LaGuardia Airport            | 94448            | 87413      | 92.55               | 69.46                | 9.41                    |
| 2  | 230                | Manhattan      | Times Sq/Theatre District    | 109175           | 11856      | 10.86               | 29.95                | 2.89                    |
| 3  | 161                | Manhattan      | Midtown Center               | 157679           | 10750      | 6.82                | 27.25                | 2.31                    |
| 4  | 236                | Manhattan      | Upper East Side North        | 155473           | 7988       | 5.14                | 22.50                | 2.03                    |
| 5  | 162                | Manhattan      | Midtown East                 | 113648           | 7824       | 6.88                | 25.98                | 2.28                    |
| 6  | 231                | Manhattan      | TriBeCa/Civic Center         | 49361            | 7388       | 14.97               | 29.79                | 3.35                    |
| 7  | 70                 | Queens         | East Elmhurst                | 8295             | 7388       | 89.07               | 65.82                | 9.10                    |
| 8  | 237                | Manhattan      | Upper East Side South        | 176330           | 7237       | 4.10                | 21.90                | 1.81                    |
| 9  | 48                 | Manhattan      | Clinton East                 | 83191            | 7224       | 8.68                | 25.89                | 2.59                    |
| 10 | 140                | Manhattan      | Lenox Hill East              | 73074            | 6875       | 9.41                | 24.97                | 2.69                    |
| 11 | 170                | Manhattan      | Murray Hill                  | 94236            | 6736       | 7.15                | 25.68                | 2.38                    |
| 12 | 163                | Manhattan      | Midtown North                | 93132            | 6685       | 7.18                | 26.55                | 2.36                    |
| 13 | 186                | Manhattan      | Penn Station/Madison Sq West | 113022           | 6317       | 5.59                | 26.42                | 2.30                    |
| 14 | 68                 | Manhattan      | East Chelsea                 | 88310            | 6306       | 7.14                | 27.68                | 2.59                    |
| 15 | 79                 | Manhattan      | East Village                 | 93298            | 5927       | 6.35                | 25.46                | 2.54                    |
| 16 | 142                | Manhattan      | Lincoln Square East          | 108049           | 5548       | 5.13                | 23.71                | 2.23                    |
| 17 | 263                | Manhattan      | Yorkville West               | 68476            | 5474       | 7.99                | 23.09                | 2.46                    |
| 18 | 75                 | Manhattan      | East Harlem South            | 35002            | 5140       | 14.68               | 24.87                | 3.29                    |
| 19 | 239                | Manhattan      | Upper West Side South        | 95862            | 5112       | 5.33                | 23.58                | 2.28                    |

## Appendix E.6 taxi\_subway\_comparison

|    | borough       | time_period  | total_taxi_trips | total_subway_ridership | taxi_to_subway_ratio |
|----|---------------|--------------|------------------|------------------------|----------------------|
| 0  | Manhattan     | Late night   | 574286           | 5060615.0              | 0.1135               |
| 1  | Manhattan     | Off peak     | 1452930          | 26296073.0             | 0.0553               |
| 2  | Queens        | Late night   | 65539            | 1322377.0              | 0.0496               |
| 3  | Manhattan     | Morning peak | 367059           | 9444184.0              | 0.0389               |
| 4  | Manhattan     | Evening peak | 802034           | 23345030.0             | 0.0344               |
| 5  | Queens        | Evening peak | 76758            | 3396125.0              | 0.0226               |
| 6  | Queens        | Off peak     | 146331           | 7064561.0              | 0.0207               |
| 7  | Brooklyn      | Late night   | 34326            | 1757674.0              | 0.0195               |
| 8  | Queens        | Morning peak | 32769            | 4761835.0              | 0.0069               |
| 9  | Bronx         | Late night   | 4831             | 768696.0               | 0.0063               |
| 10 | Brooklyn      | Off peak     | 59996            | 10672219.0             | 0.0056               |
| 11 | Bronx         | Off peak     | 16573            | 3375027.0              | 0.0049               |
| 12 | Bronx         | Morning peak | 10003            | 2115278.0              | 0.0047               |
| 13 | Brooklyn      | Morning peak | 27923            | 6860317.0              | 0.0041               |
| 14 | Staten Island | Late night   | 51               | 13907.0                | 0.0037               |
| 15 | Brooklyn      | Evening peak | 19606            | 6116235.0              | 0.0032               |
| 16 | Bronx         | Evening peak | 4678             | 1494566.0              | 0.0031               |
| 17 | Staten Island | Off peak     | 91               | 63704.0                | 0.0014               |
| 18 | Staten Island | Morning peak | 65               | 51629.0                | 0.0013               |
| 19 | Staten Island | Evening peak | 53               | 63912.0                | 0.0008               |

## Appendix E.7 taxi\_bus\_comparison

|    | borough       | time_period  | total_taxi_trips | total_bus_ridership | taxi_to_bus_ratio |
|----|---------------|--------------|------------------|---------------------|-------------------|
| 0  | Manhattan     | Late night   | 574,286          | 303,206             | 1.8940            |
| 1  | Manhattan     | Off peak     | 1,452,930        | 3,308,650           | 0.4391            |
| 2  | Manhattan     | Evening peak | 802,034          | 2,028,627           | 0.3954            |
| 3  | Manhattan     | Morning peak | 367,059          | 1,458,824           | 0.2516            |
| 4  | Brooklyn      | Late night   | 34,326           | 405,625             | 0.0846            |
| 5  | Queens        | Late night   | 65,539           | 849,662             | 0.0771            |
| 6  | Queens        | Off peak     | 146,331          | 4,962,038           | 0.0295            |
| 7  | Queens        | Evening peak | 76,758           | 2,999,247           | 0.0256            |
| 8  | Brooklyn      | Off peak     | 59,996           | 3,438,223           | 0.0174            |
| 9  | Brooklyn      | Morning peak | 27,923           | 1,681,088           | 0.0166            |
| 10 | Queens        | Morning peak | 32,769           | 2,470,113           | 0.0133            |
| 11 | Brooklyn      | Evening peak | 19,606           | 2,003,286           | 0.0098            |
| 12 | Staten Island | Late night   | 51               | 173,267             | 0.0003            |
| 13 | Staten Island | Morning peak | 65               | 398,983             | 0.0002            |
| 14 | Staten Island | Off peak     | 91               | 729,209             | 0.0001            |
| 15 | Staten Island | Evening peak | 53               | 503,480             | 0.0001            |
| 16 | Bronx         | Off peak     | 16,573           | 0                   | nan               |
| 17 | Bronx         | Evening peak | 4,678            | 0                   | nan               |
| 18 | Bronx         | Morning peak | 10,003           | 0                   | nan               |
| 19 | Bronx         | Late night   | 4,831            | 0                   | nan               |

## Appendix E.8 taxi\_bike\_hourly\_comparison

|    | hour | short_taxi_trips | bike_trips |
|----|------|------------------|------------|
| 0  | 0    | 44360            | 42878      |
| 1  | 1    | 29578            | 24992      |
| 2  | 2    | 18937            | 16044      |
| 3  | 3    | 12094            | 10678      |
| 4  | 4    | 8083             | 11004      |
| 5  | 5    | 9623             | 26723      |
| 6  | 6    | 23327            | 78291      |
| 7  | 7    | 53743            | 163338     |
| 8  | 8    | 80195            | 261711     |
| 9  | 9    | 88168            | 212965     |
| 10 | 10   | 90284            | 173795     |
| 11 | 11   | 99650            | 181183     |
| 12 | 12   | 109280           | 201312     |
| 13 | 13   | 111143           | 212404     |
| 14 | 14   | 120635           | 225718     |
| 15 | 15   | 125306           | 250416     |
| 16 | 16   | 123287           | 286066     |
| 17 | 17   | 139358           | 382080     |
| 18 | 18   | 143492           | 358291     |
| 19 | 19   | 122231           | 264597     |
| 20 | 20   | 109533           | 173796     |
| 21 | 21   | 110163           | 129035     |
| 22 | 22   | 98965            | 102133     |

---

# Appendix F: Data Ethics and Governance Checkpoints

## Appendix F.1 Data Minimisation Log

| dataset          | fields_used                                                                                                | fields_avoided_or_limited                                                                                                                             | reason                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Taxi             | pickup/dropoff datetime, pickup/dropoff location ID, borough, zone, trip distance, total amount, taxi type | vendor ID, payment type, passenger count for core demand analysis                                                                                     | The analysis focuses on taxi demand by time, location, distance, and intervention priority. Passenger count has quality issues and is not used as the main demand metric. |
| Taxi zone lookup | LocationID, borough, zone, service zone                                                                    | None beyond mapping fields                                                                                                                            | Only required for joining location IDs to readable geographic areas.                                                                                                      |
| Subway           | transit timestamp, borough, station complex, ridership, transfers                                          | Individual passenger information is not available or used                                                                                             | Used only as aggregated public transport context by borough and time period.                                                                                              |
| Bus              | transit timestamp, bus route, route prefix, ridership, transfers                                           | Payment method and fare class are not used for behavioural claims                                                                                     | Used only as broad route-level and time-period public transport context.                                                                                                  |
| Citi Bike        | ride ID, start/end time, station names, coordinates, bike type, user type                                  | No individual ride paths are visualised; station IDs are treated as text identifiers; coordinates are used only for aggregated station/spatial checks | Used as aggregated context for assessing whether Citi Bike may support short-trip taxi alternatives, without analysing individual-level movement patterns.                |

## Appendix F.2 Aggregation and Privacy Control Log

| analysis_output                                | aggregation_level               | privacy_control                                                             | why_it_matters                                                                                                            |
|------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Taxi demand by borough                         | Borough                         | Aggregates many trips into borough-level totals                             | Avoids showing individual pickup/dropoff movements                                                                        |
| Taxi demand by pickup hour                     | Hourly summary                  | Uses total trips by hour rather than individual timestamps                  | Reduces risk of identifying repeated travel behaviour                                                                     |
| Taxi distance pattern by borough               | Borough + distance category     | Uses distance categories instead of individual trip distances               | Supports strategic interpretation without exposing specific trips                                                         |
| Taxi-to-subway ratio                           | Borough + time period           | Compares aggregate taxi trips and aggregate subway ridership                | Avoids individual-level public transport inference                                                                        |
| Taxi-to-bus ratio                              | Borough proxy + time period     | Uses aggregate bus ridership and taxi trips                                 | Provides context without claiming individual mode substitution                                                            |
| Taxi pickup volume map                         | Taxi zone                       | Maps zone-level volume categories rather than raw pickup points             | Reduces spatial precision and avoids trip-level mapping                                                                   |
| Short taxi trips vs Citi Bike activity by hour | Hourly summary                  | Compares aggregated hourly counts rather than individual taxi or bike trips | Supports short-trip alternative analysis without showing individual ride paths, exact trips, or repeated travel behaviour |
| Recommended intervention map                   | Taxi zone intervention category | Shows policy-relevant categories, not individual trip records               | Supports planning while avoiding individual mobility disclosure                                                           |

## Appendix F.3 Responsible Wording Log

| analysis_signal                         | avoid_claiming                                     | use_instead                                                                                               |
|-----------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| High taxi demand in a borough or zone   | People in this area rely on taxis more than others | This area shows high recorded taxi pickup activity and may require operational review                     |
| High taxi-to-subway ratio               | Subway service is poor or unavailable              | Taxi use is high relative to subway ridership, suggesting a public transport context worth reviewing      |
| High late-night taxi activity           | Residents have no safe late-night transport        | Late-night taxi demand is elevated, which may justify reviewing late-night mobility options               |
| Short taxi trips concentrated in a zone | People should switch to bikes or buses             | Short-trip concentration suggests potential for investigating alternative short-distance mobility options |
| Low taxi activity                       | This area has low mobility need                    | Recorded taxi activity is low; this does not capture unmet demand or non-taxi travel needs                |

# Appendix G: Taxi Demand Overview Outputs

## Appendix G.1 Borough-Level Taxi Summary

|   | pickup_borough | total_taxi_trips | yellow_taxi_trips | green_taxi_trips |
|---|----------------|------------------|-------------------|------------------|
| 0 | Manhattan      | 3196309          | 3170897           | 25412            |
| 1 | Queens         | 321397           | 312601            | 8796             |
| 2 | Brooklyn       | 141851           | 135172            | 6679             |
| 3 | Bronx          | 36085            | 34890             | 1195             |
| 4 | Staten Island  | 260              | 252               | 8                |

## Appendix G.2 Pickup Zone Summary and Ranking

|   | zone_label                            | pickup_borough | pickup_zone               | total_taxi_trips | priority_level  | recommended_intervention        | priority_score | signal_count | signal_summary                                    | interpretation_noti                              |
|---|---------------------------------------|----------------|---------------------------|------------------|-----------------|---------------------------------|----------------|--------------|---------------------------------------------------|--------------------------------------------------|
| 0 | Manhattan - Times Sq/Theatre District | Manhattan      | Times Sq/Theatre District | 109175           | High priority   | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi ... | Candidate for curbside pickup/drop-of managem... |
| 1 | Manhattan - East Village              | Manhattan      | East Village              | 93298            | High priority   | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi ... | Candidate for curbside pickup/drop-of managem... |
| 2 | Manhattan - Clinton East              | Manhattan      | Clinton East              | 83191            | High priority   | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi ... | Candidate for curbside pickup/drop-of managem... |
| 3 | Manhattan - West Village              | Manhattan      | West Village              | 80404            | High priority   | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi ... | Candidate for curbside pickup/drop-of managem... |
| 4 | Manhattan - Flatiron                  | Manhattan      | Flatiron                  | 53384            | High priority   | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi ... | Candidate for curbside pickup/drop-of managem... |
| 5 | Manhattan - Greenwich Village South   | Manhattan      | Greenwich Village South   | 53280            | High priority   | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi ... | Candidate for curbside pickup/drop-of managem... |
| 6 | Manhattan - Greenwich Village North   | Manhattan      | Greenwich Village North   | 52590            | High priority   | Taxi pickup/drop-off management | 5.0            | 3            | High taxi pickup volume; High late-night taxi ... | Candidate for curbside pickup/drop-of managem... |
| 7 | Manhattan - Upper East Side South     | Manhattan      | Upper East Side South     | 176330           | Medium priority | Taxi pickup/drop-off management | 3.5            | 2            | High taxi pickup volume; High short-trip conce... | Candidate for curbside pickup/drop-of managem... |
| 8 | Manhattan - Midtown Center            | Manhattan      | Midtown Center            | 157679           | Medium priority | Taxi pickup/drop-off management | 3.5            | 2            | High taxi pickup volume; High short-trip conce... | Candidate for curbside pickup/drop-of managem... |
| 9 | Manhattan - Upper East Side North     | Manhattan      | Upper East Side North     | 155473           | Medium priority | Taxi pickup/drop-off management | 3.5            | 2            | High taxi pickup volume; High short-trip conce... | Candidate for curbside pickup/drop-of managem... |

## Appendix G.3 Taxi Hourly Demand Summary

| index | pickup_hour | total_taxi_trips |
|-------|-------------|------------------|
| 0     | 0           | 105792           |
| 1     | 1           | 67700            |
| 2     | 2           | 43702            |
| 3     | 3           | 30329            |
| 4     | 4           | 25240            |
| 5     | 5           | 31393            |
| 6     | 6           | 64557            |
| 7     | 7           | 117628           |
| 8     | 8           | 156591           |
| 9     | 9           | 163600           |
| 10    | 10          | 163896           |
| 11    | 11          | 177841           |
| 12    | 12          | 192323           |
| 13    | 13          | 198397           |
| 14    | 14          | 217494           |
| 15    | 15          | 221703           |
| 16    | 16          | 215235           |
| 17    | 17          | 236087           |
| 18    | 18          | 238932           |
| 19    | 19          | 212875           |
| 20    | 20          | 213067           |
| 21    | 21          | 226643           |
| 22    | 22          | 213128           |
| 23    | 23          | 161749           |



# Appendix H: Detailed Taxi Pattern Outputs

## Appendix H.1 Detailed Distance-Pattern Outputs

| trip_distance_category | Short trip | Medium trip | Long trip |
|------------------------|------------|-------------|-----------|
| pickup_borough         |            |             |           |
| Bronx                  | 10.49      | 29.91       | 59.60     |
| Brooklyn               | 16.48      | 46.74       | 36.79     |
| Manhattan              | 58.58      | 33.39       | 8.03      |
| Queens                 | 6.10       | 14.63       | 79.27     |
| Staten Island          | 12.31      | 26.15       | 61.54     |

## Appendix H.2 Detailed Late-Night Zone Outputs

|    | pickup_location_id | pickup_borough | pickup_zone                  | total_taxi_trips | late_night_trips | late_night_trip_share_pct | avg_total_amount_usd | avg_trip_distance_miles |
|----|--------------------|----------------|------------------------------|------------------|------------------|---------------------------|----------------------|-------------------------|
| 0  | 79                 | Manhattan      | East Village                 | 93298            | 42563            | 45.62                     | 25.46                | 2.54                    |
| 1  | 249                | Manhattan      | West Village                 | 80404            | 33202            | 41.29                     | 25.54                | 2.29                    |
| 2  | 132                | Queens         | JFK Airport                  | 135796           | 29621            | 21.81                     | 77.78                | 14.79                   |
| 3  | 230                | Manhattan      | Times Sq/Theatre District    | 109175           | 25779            | 23.61                     | 29.95                | 2.89                    |
| 4  | 148                | Manhattan      | Lower East Side              | 45926            | 25621            | 55.79                     | 28.40                | 3.08                    |
| 5  | 114                | Manhattan      | Greenwich Village South      | 53280            | 25200            | 47.30                     | 25.82                | 2.37                    |
| 6  | 48                 | Manhattan      | Clinton East                 | 83191            | 24424            | 29.36                     | 25.89                | 2.59                    |
| 7  | 161                | Manhattan      | Midtown Center               | 157679           | 19378            | 12.29                     | 27.25                | 2.31                    |
| 8  | 186                | Manhattan      | Penn Station/Madison Sq West | 113022           | 18819            | 16.65                     | 26.42                | 2.30                    |
| 9  | 142                | Manhattan      | Lincoln Square East          | 108049           | 18379            | 17.01                     | 23.71                | 2.23                    |
| 10 | 234                | Manhattan      | Union Sq                     | 93684            | 18195            | 19.42                     | 25.04                | 2.14                    |
| 11 | 68                 | Manhattan      | East Chelsea                 | 88310            | 16997            | 19.25                     | 27.68                | 2.59                    |
| 12 | 138                | Queens         | LaGuardia Airport            | 94448            | 16246            | 17.20                     | 69.46                | 9.41                    |
| 13 | 164                | Manhattan      | Midtown South                | 70043            | 16033            | 22.89                     | 26.36                | 2.41                    |
| 14 | 237                | Manhattan      | Upper East Side South        | 176330           | 14929            | 8.47                      | 21.90                | 1.81                    |
| 15 | 163                | Manhattan      | Midtown North                | 93132            | 14321            | 15.38                     | 26.55                | 2.36                    |
| 16 | 144                | Manhattan      | Little Italy/NoLiTa          | 38915            | 14041            | 36.08                     | 27.43                | 2.60                    |
| 17 | 107                | Manhattan      | Gramercy                     | 67305            | 13912            | 20.67                     | 24.89                | 2.37                    |
| 18 | 113                | Manhattan      | Greenwich Village North      | 52590            | 12772            | 24.29                     | 25.21                | 2.21                    |
| 19 | 90                 | Manhattan      | Flatiron                     | 53384            | 12751            | 23.89                     | 24.76                | 2.24                    |